Agency for Adaptive Architecture


i06 Instrumentet 17


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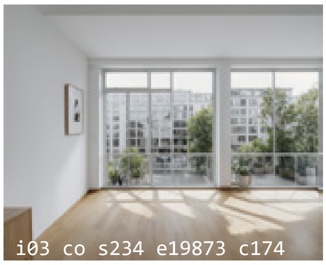
Could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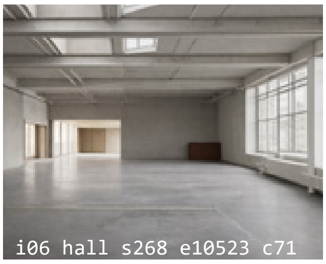
You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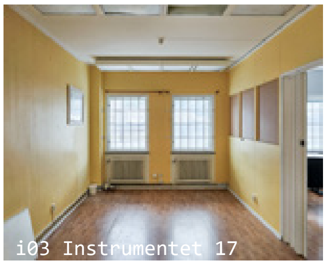
Live


i03 Instrumentet 17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Here?

couldyoulivehere.dev


i03 apt s028 e20458 c227


i06 hall s116 e8052 c50

Vacant Potential

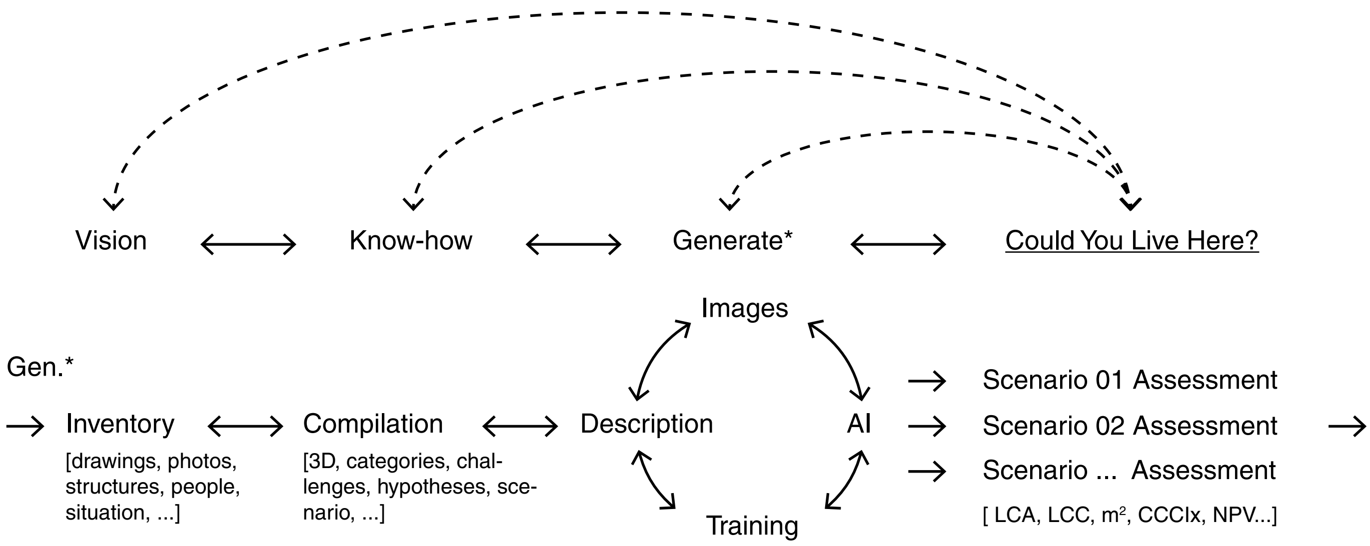
The way we live and work in the city is rapidly changing. Some buildings are having a hard time to adapt, not being agile enough to respond to new needs, the expectations from the surrounding city or solving its own technical problems. Within our current models and practices, they are deemed out, but if we take a closer look, could they be hiding a vacant potential?

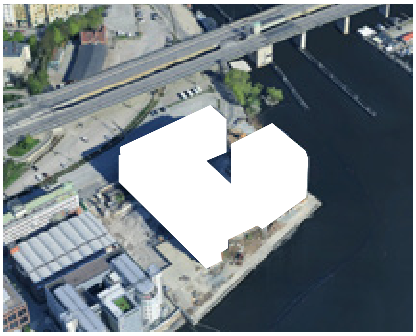
This work has closely followed the progression on how the building and planning sector wants to work (differently) with the existing environment during the last couple of years. I am sure it has passed no one unnoticed the discussion and large leaps made, from friction and conflict to new legislation, new practices and new visions for how we build our cities. It is those clear that we* (the planning and building sector) should take more care of our built legacy and resources, but how this is done has yet many questions to be answered.

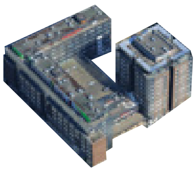
How planning and building was done and envisioned before continues to have an impact on what is happening today. Things take time. Decisions are carefully considered, but also meaning, many decisions are very hard to take back. Looking forward, from today, the challenge lies in identifying and put into use the possibilities that fits within the existing framework of a building. This also means negotiating how the logic of a building fits the logic of a property.

In following several properties, the shift in both how our process in working with the existing environment, but also, the understanding of how this can be done, is becoming clearer. As a small part of a whole new thinking, a tailored AI workflow visualizes and evaluates the possibilities that lies within an existing framework, later referred to as Generative Scenario Assessment. In collaboration with the developer Riksbyggen, an in depth pilot has been made on the property Instrumentet 17 in Örnsberg, Stockholm.

* When discussing change, it is easy to talk about a We, which implicates a Them. Change is by some welcomed by open arms, other more reluctant and some work against it. Be it fuelled by private interest, moral conviction, cultural and aesthetic preferences, habit, logical determination or opportunism, the overall picture is complex and fragmented. The We, implicates an agency, or the decision for it.

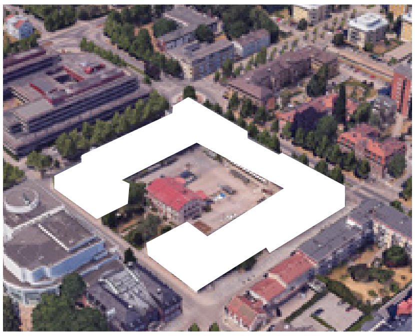

Vision
Know-how
Generate*
Could You Live Here?
Images
Gen.*
Inventory
[drawings, photos, structures, people, situation, ...]
Compilation
[3D, categories, challenges, hypotheses, scenario, ...]
Description
AI
Training
Scenario 01 Assessment
Scenario 02 Assessment
Scenario ... Assessment
[LCA, LCC, m², CCCIx, NPV...]

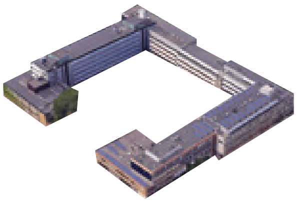




<u>Marievik 15</u>

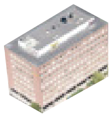
Liljeholmen, Stockholm. A large office complex developed mainly during the late 1980s century, with a distinctive architectural identity and a prominent position along the waterfront. After becoming largely vacant, the property became the centre of a wider discussion around demolition and the potential reuse of existing office buildings. Demolition soon began when the debate intensified.





<u>Ansgarius 15</u>

Växjö. A city block that had grown through several additions over many decades and previously accommodated municipal offices. When the municipality moved to a newly built city hall by the railway station, the existing property was intended for conversion into housing. This proved difficult, and the building remained largely vacant for several years before being sold.



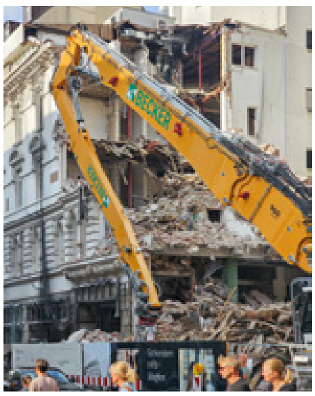
<u>Primus 1</u>

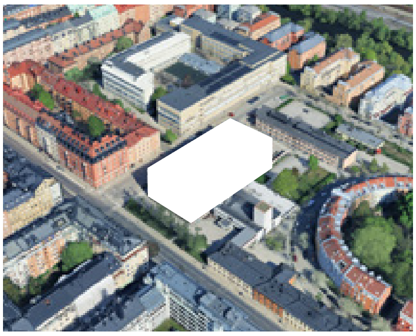
Lilla Essingen, Stockholm. An office and industrial property forming part of a larger redevelopment area on the northern part of the island. The planning process went through several iterations, including proposals for partial preservation and a fan-shaped urban structure. As economic conditions changed, renewed calls were made to retain parts of the existing complex. By then, however, the planning process had progressed too far, and a building that could have fitted relatively precisely within the dimensions of the new development was demolished.

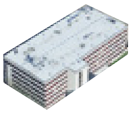
<u>None Agile Buildings</u>

The shift in how we consider existing buildings as a resource in the development of the city has taken shape through following several properties and urban planning schemes since 2023. The practice and idea of preservation is expanding, giving reason and voice to buildings that might previously have been easily dismissed.

Can a building have its own agency – does it want something – and, by extension, what can it give back?

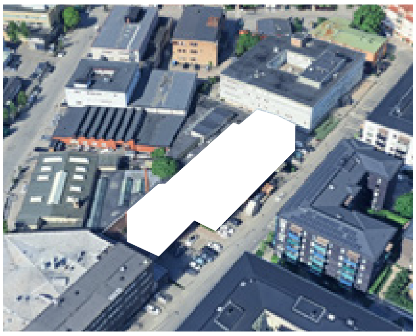


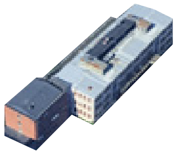




<u>Tegelbruket 4</u>

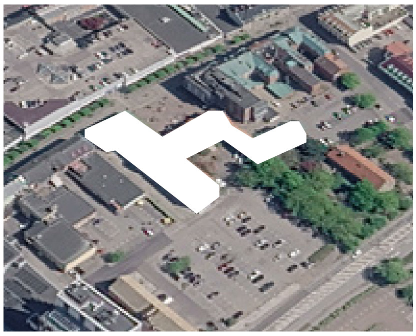
Stockholm. The former St Erik Eye Hospital, centrally located in Kungsholmen, was sold by Region Stockholm to a private developer as part of a larger redevelopment for housing. During the planning process, different degrees of reuse and partial preservation were investigated. Reuse appeared economically feasible at an early stage, but would have required changes to the urban plan and to the vision of the traditional “stone city”.

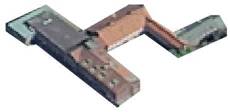




<u>Instrumentet 17</u>

Örnsberg, Stockholm. An office and light-industrial property originally planned for demolition as part of a future redevelopment. In collaboration with Riksbyggen, the building became the main pilot case for testing the developed assessment workflow. Because the planning process had not progressed as far as in several of the other cases, demolition could still be reconsidered and alternative scenarios investigated.





<u>Herkules 31</u>

Trelleborg. The former town hall complex, which has served as an administrative and political centre of the city since the mid-19th century. As in Växjö, the municipality moved from its own property into newly built premises under a rental arrangement. The existing buildings now stand largely vacant, facing both the complexity of adaptation and interest. As many institutional buildings from the time of the well fare state expansion, as they are “ugly”, they are not fitting the political vision of the city development.

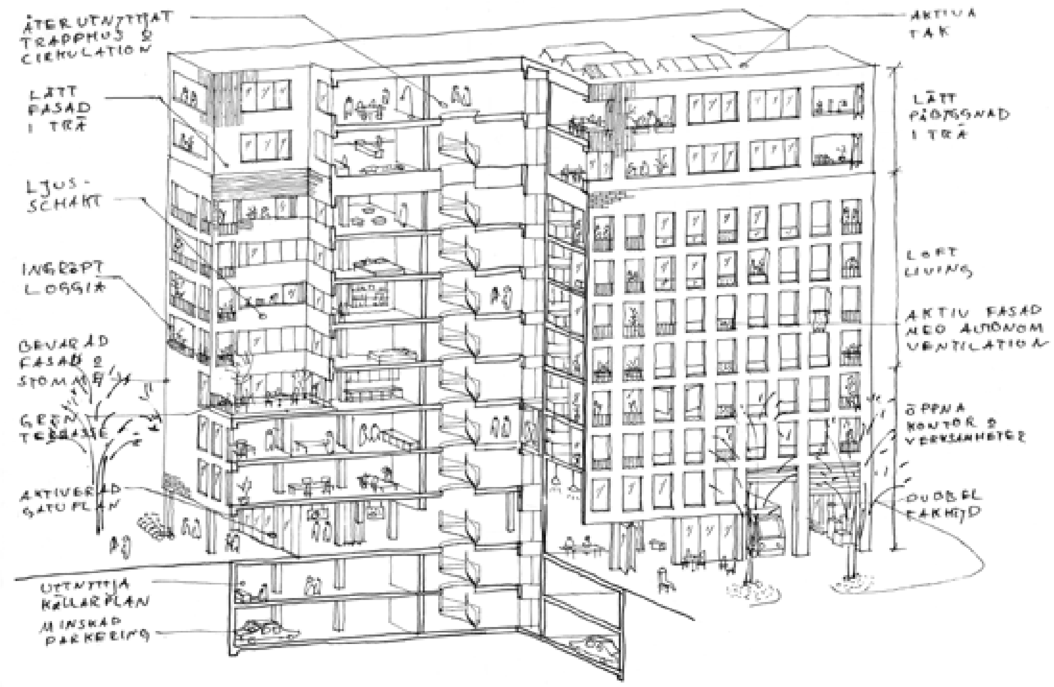
Building with the Legacy of Yesterday

When we speak about the existing environment, there is a physical reality that connects rooms with people, and functions with the city. It's a complex web of aspects, some you can measure, others equally important but counterproductive to put a number to. Weighing them against each other thus becomes a difficult task, also demanding different methods and tools. The composition of materials, the structure behind it, the history of that brick, the many people who have created memories, the dimensions between the pillars and of the windows allowing light and space, the ceiling height allowing people to be, the ventilation system or the lack of it, the CO_2 embodied in the concrete and linoleum flooring, and, last but not least, the economy: budget, expectation, projection.

These different values do not stop at the property boundary. What happens to one building also affects the larger structure of the city. Housing is central, partly because it is the most common typology, and partly because voices repeatedly call for a debate, or point to the lack of debate, around the role of housing in urban development. Everything is connected, but the value of this physical reality can stand in contrast to the possible future of a property on paper. The economic potential of a development right, or of a potential future, is weighed against the building that already stands on the site. There is an expectation that the city should develop and renew itself, both to respond to new or changing needs and to create more value.

Urban planning is increasingly complex, simply because we have higher demands and diverse uses that should come together without affecting each other in unwanted ways. When sitting around the planning table, how could this former office building be part of a new housing development? It is tailored to another specific use and coloured by its time. This leads us directly into the concrete reality of the building that stands there, blurring the lines between architecture and urban planning. The complexity increases. Where we could before follow a straighter line, working with the existing environment requires us to jump between different scales and materials, and to start a dialogue between the actors concerned from an early stage.

The question is therefore not only what values an existing building contains, but which of those values our current systems are able to recognise, measure and act upon. This is challenging current practice. In Sweden, there is no comprehensive statistic on how many properties are



converted or demolished. To demolish a building and construct a new one on the same site requires more resources and breaks a link to history and culture. People's memories and activities are moved or disappear. Through the municipalities' planning monopoly, where preservation can be marked in the detailed development plan, and through the requirement to apply for a demolition permit, there is a certain protection for the existing environment. This, however, is mostly done in relation to cultural-historical values, and if a property owner is denied demolition, for example where the detailed development plan allows for a larger development, they may in some cases be financially compensated.

An important detail is that the tools we use today to calculate a building's climate impact, through Life Cycle Assessment, LCA, depend on what is included within the calculation. Ground preparation, previous demolition, or the fact that an existing building already stands on the site are not always included in a way that makes reuse and new construction directly comparable. This means that a building can be demolished and a new one built that still achieves the highest environmental certifications. A newly built timber structure can even appear to 'store' more carbon dioxide than an existing building, while the emissions from constructing the existing building have already happened and do not appear again in the calculation. Demolition and new construction can therefore, depending on how the calculation is made, appear to add up against renovation.

The task of a property owner is to manage and develop their property in the best possible way. Based on how economic potential is calculated today, it can often appear that demolishing and building new creates the greatest economic value, for the property owner, but also, by extension, for those who use or buy a home. Economic value is itself dependent on what is counted, over what period, under which rules, and for whom. That is not to say that there are not values that are difficult to measure monetarily, or that the legislation, process and vision that fundamentally govern what we build today are not optimised for new construction. If we discussed these from the perspective of the existing, another kind of city, architecture, housing and even economy would take shape.

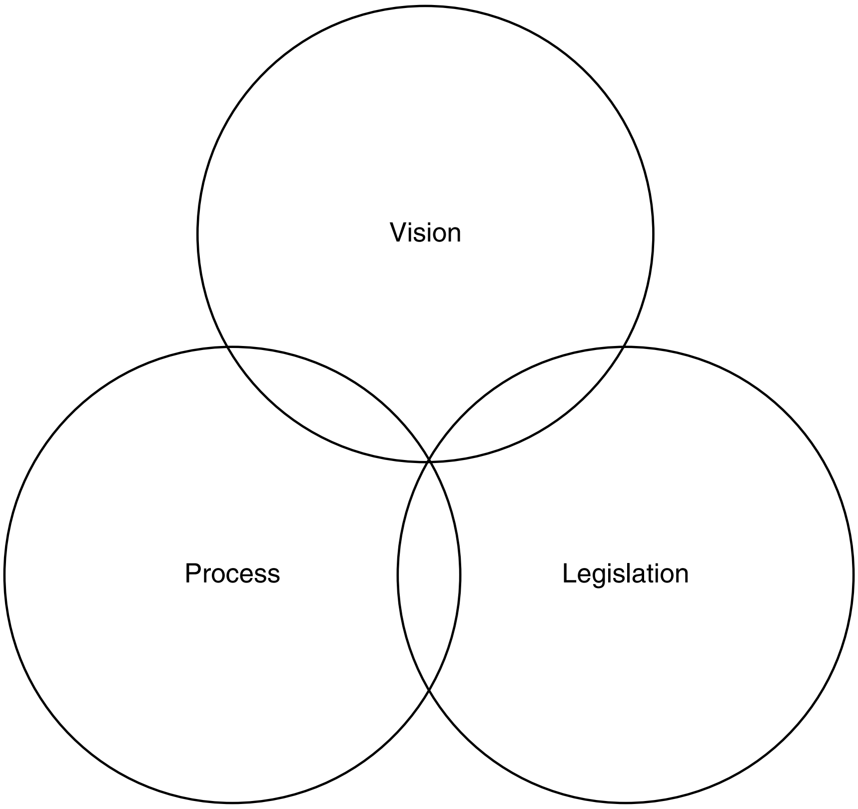
Legislation, Process, Vision

Risk appears when a process enters the unknown, when there is uncertainty about how different steps should be carried out and calculated. The existing building introduces risk. To develop and build new is more predictable. The process around a new building is a well-oiled machine that has constantly been optimised to reduce costs. The building frame, the basic code of our buildings today, still owes much to the industrialised construction that developed strongly during the Swedish Million Programme. By contrast, all work with the existing is specific, lacking standardisation or classification. The experience that exists around conversion and adaptation is neither gathered nor organised in the same way as it is for building new.

Property development is one of the most capital-intensive sectors of the economy. Construction is expensive, planning processes are lengthy and uncertain, and real estate accounts for a substantial share of national wealth and bank lending. Minimising risk is thus a key concern, often meaning knowing as much as possible. With knowledge, understanding and experience, we can predict the outcome, cost, profit and impact, of a proposal. But when faced with an existing building, it is often not clear where to start. No up-to-date drawings or no one knows what is hiding inside the walls, nor is there a general understanding of what the structure and space could host.

In many ways, it is a move from the abstract to the concrete. In an optimised process of planning and building hors-situ, the drawings and spreadsheets are rarely incorrect, they are mostly dependent on each other. But now, when approaching a structure and material that are already there, making sense as they are, there is so much we do not know about their properties, their context and their possibilities. If building new is operating within the predictable, working with the existing needs to introduce design thinking earlier in the process to expand the frames of the operation. This means leaving linearity for exploration.

If we ask a thick building like Marievik 15, can you be standard apartments, the answer will be no. But if the question is formulated differently, can you accommodate living spaces, or could you be a mix of living spaces and working spaces, the answer might be something else. In between the physical reality, the regulatory framework and feasibility, there is a cyclical matching of typologies. Different buildings, and different parts of the same building, can have a multitude of responses, potentially even giving back more than you hoped for. An overlooked quality of a space can be the aspect that lifts a certain use and renovation to another level. Expanding the scope beyond standard apartments to different forms of living opens up the potential to find a well-matched typology



where the existing building gives something more and unexpected back.

Here there is also a strong shift. Traditionally, we have measured the worth of an existing building through its legacy, its historical and cultural layers. Today, however, an existing building is more than that. It is key to building sustainable and resilient cities.

It is a resource, to be dealt with care and efficiency.

In this respect, the architect has an important role. Closely interwoven with economy, technology and ecology, it is a matter of seeing possibilities in the complex reality that an existing environment entails. Working with the existing means moving between three interconnected layers: legislation, process and vision. They change more or less easy at different speeds, but continually shape one another.

Legislation. It provides the rules within which a property can be developed: how a dwelling can be designed, and what requirements apply when changes are made to an existing building. Legislation is both protection and constraint. Building regulations and technical standards are still largely interpreted through the logic of new construction, creating friction when applied to existing structures. At the same time, there is room for interpretation that is not always fully used. Today, these rules are under scrutiny, with new laws and government initiatives aimed at making renovation and conversion easier.

Process. Urban development is governed by many different processes, which together create a kind of culture around how things are done. It is about how we build houses, how financing models are structured, and in what order different decisions are made. For existing buildings, uncertainty can too easily be understood as an obstacle rather than as knowledge to work with. Interwoven with legislation, different perspectives are valued and create a pattern for how the city develops.

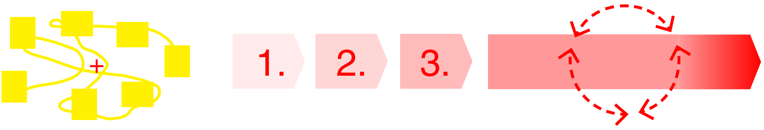
During the planning process, there are no public documents that show the weighing of risk and potential in the existing building stock. Risks and cultural-historical values are evaluated, but the property owner's analysis of the economic possibilities of the existing building is not available to the public. This is one reason why the question of demolition often becomes difficult to understand and, to some extent, politicised. It is the property owner who investigates how the property should be developed. On the one hand, demolition becomes a discussion about competing visions. On the other, it reveals an imbalance of knowledge and a need to place the question in perspective. Emerging from the climate question, and tied to ideas about the good city, the debate seeks to find new answers and increase knowledge about the existing not only as risk, but also as potential.

In autumn 2024, Boverket arranged a seminar day in Växjö titled Idéverkstad omvandling. Perspectives were gathered broadly from the industry on how the difficulties of conversion could be better understood. A local example was raised: Växjö's old municipal building, Ansgarius 15, had stood empty for several years. After the municipality's activities moved to a new building, the existing building was intended to be converted into housing. But in the borderland between administrative processes, specific difficulties in relation to the detailed development plan, and the requirements of housing, the project had still not been possible to carry out. If these frameworks change, both the possibilities and the dwelling are redrawn. Other typologies may be approved, bringing other qualities and norms, which in turn determine whether a conversion is profitable or not.

Ansgarius 15 has, after being empty for many years, now been sold to a private developer, with offices, hotel and potentially housing as possible uses. The property stood empty for a long time while the municipality focused on converting the whole urban block into housing. Seen from this perspective, the question appears to have been too narrowly formulated, the internal process at the municipality suffering from silo-thinking in different parts of the organisation. The building was sold at an exceptionally low price, arguably a loss for the public, but also a missed opportunity to push forward new methods and processes for how such a building can find new use.

Vision. In the borderland between different perspectives, it becomes all the clearer when we open a newspaper: as a society, we want to take better care of what we already have. From the perspective of climate, culture and a more long-term economy, this direction is becoming clearer. Vision expresses values and possible futures, but it quickly meets demands for measurability, feasibility and regulatory compliance. When reduced to programme requirements or key figures, it risks losing its spatial and social dimension. With different wills pulling in different directions, it is nevertheless clear that the debate is on the agenda.

Legislation is what society has decided is allowed, process is how and with whom decisions are made, and vision is what we can imagine, what we desire. In change, if there is the wish for it, even if it moves more slowly, legislation will follow as new visions grow stronger and our processes slowly adapt, finding new methods and tools to realise them.



Unlocking Existing Values

Design Thinking Before Design

Working with existing buildings requires the direction of a project to be explored before form and programme are fixed. Before investment decisions and visions take over, the concrete reality needs to be understood: the site, the building, its materiality, dimensions and the spatial relationships that already structure what is possible. This is where the project's actual room for manoeuvre is defined.

Unlike new construction, where standardised systems and known solutions make it possible to work abstractly far into the process, repurposing is always specific. Every building has its own logic, constraints and untapped qualities. If these are not analysed early, the project risks being driven by general assumptions rather than by the actual conditions of the site and building.

Often it is something unexpected that changes the direction. A window rhythm, the quality of the concrete when it was built, the junction of the floor slabs or the placement of the shafts can make one use feel wrong and another possible. What first appears as a limitation can become the reason to work differently. The building starts to answer back, by narrowing and opening possibilities at the same time.

By moving between spatial assessment and economic testing at an early stage, several possible directions can remain open at the same time. Technical interventions, cost, climate impact and use are tested in parallel as scenarios, rather than as finished proposals. The question is then not what the project should look like, but what the building is best suited to become.

Not every building is suited to the same programme. In some cases housing is the right direction; in others, commercial uses, offices or hybrid programmes may create greater long-term value. Testing these alternatives before design takes over reduces uncertainty and allows the design process to begin with a direction rather than a guess.

Unlocking existing values is about embracing the complexity of a building or place.

By starting from what is already there, we can create more value, continuity, diversity and craftmanship, with less material, investment, monotony and impact.

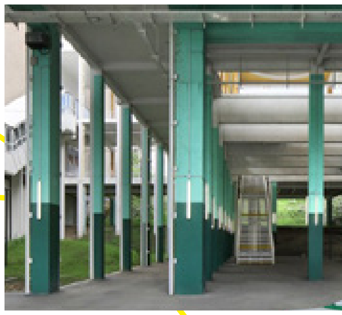
New *[neubau]* Question | Design – answer

Need ⟶ Current state ⟶ Feasability ⟶ Project ⟶ Implementation

Repurpose *[umbau]* Design Thinking

Need ⟶ Mapping ⟶ Scenarios ⟶ Project ⟶ Implementation

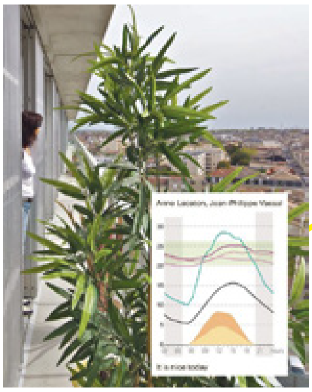
1. 2. 3.



Structure and techni

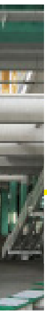
Resources and climate

Value Analysis of the



Function and generality

Economy anc



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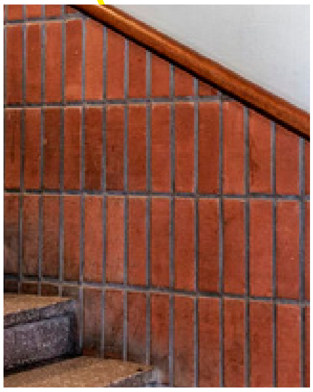
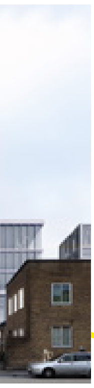
History and memory



Space and material

Existing Environment

Social and cultural



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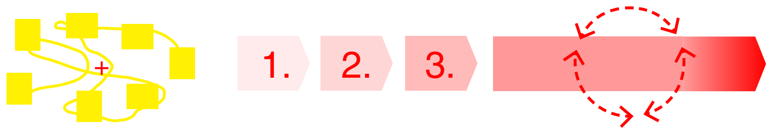
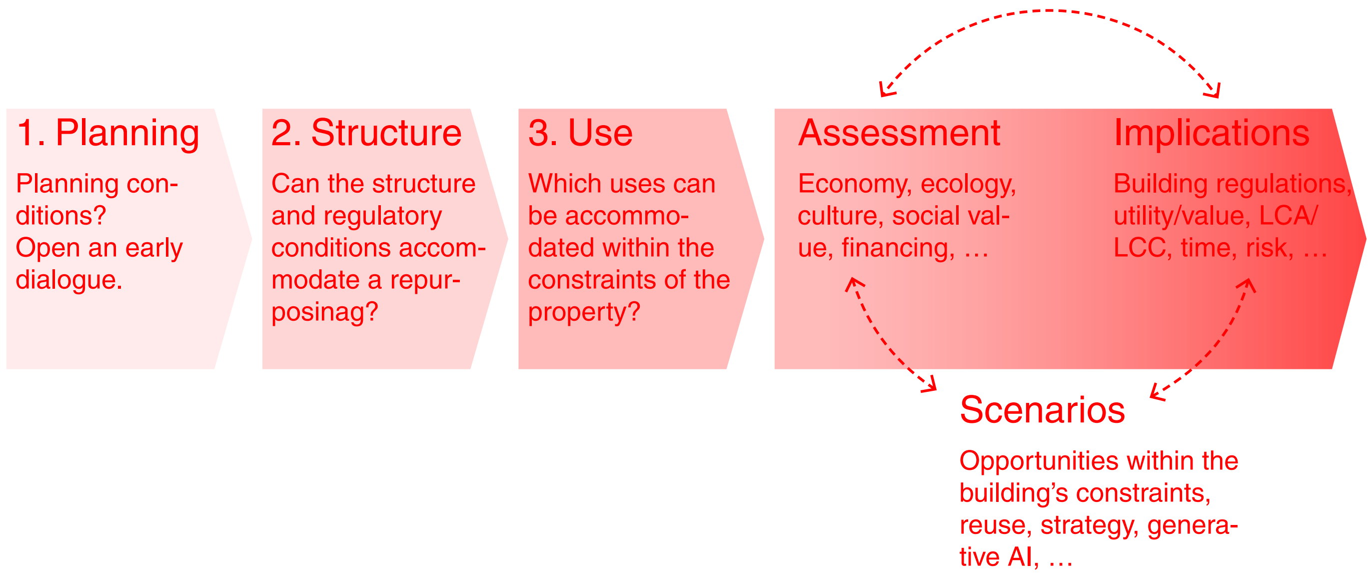
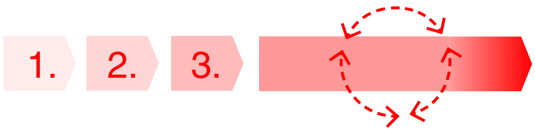


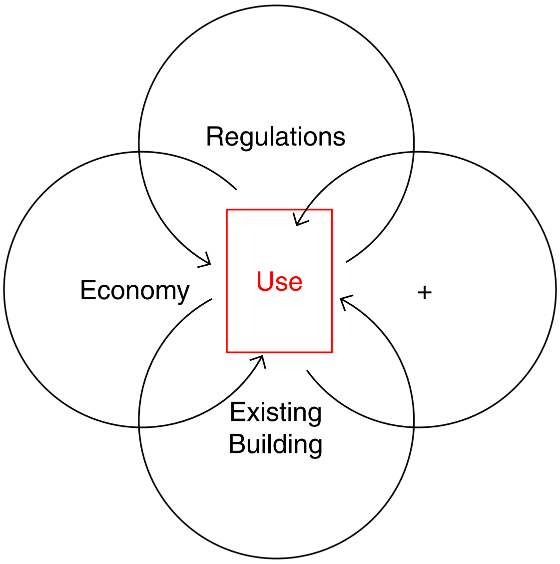
Repurpose *[umbau]*

Need →	Mapping →	Scenarios →	Project →	Implementation
Change, drivers, goals, long term perspective, ...	1. Planning 2. Structure 3. Use / group + Value analysis	Value, risk, feasa-bility, regulations, opportunities, time	Programme, plan, financing, design,	Detailed design, op-eration, construction, learning, follow-up, …
	[understand constraints]	[identify possibilities]		



Moving through the mapping process in three distinct steps makes sure that major constraints are identified early rather than discovered later. Planning conditions, structure and possible uses are tested in sequence, before scenarios are assessed and their wider implications understood.





It is a constant exchange between different layers of a building. As more knowledge is found, a chain of effects recalibrates what is feasible. Slowly moving towards a direction, the matching of typology with the existing building is at the core of making the most of what is there and creating the most value.

Renovation

Renovation
+ Addition

Deep Renovation

Deep Renovation
+ Addition

Building New

As with all existing buildings, each case is unique and hard to predict. The answer never lies in a dogma of *tabula rasa* or *tabula plena*, but in the tailored negotiation between old and new.

3.

Adaptive Typologies and New User Groups

New forms of living and working can unlock value in offices, industrial buildings and other existing structures when the programme is developed in dialogue with what is already there.

Loft Living

A housing type that accepts the existing large-scale spatial conditions and allows the space itself to remain primary. The programme is introduced as freestanding elements rather than through conventional subdivision into rooms.

Raw Living / Klusfalt

An unfinished residential framework in which the structure, services and basic functions are provided, while residents shape the finishes and spatial layout themselves. This reduces the initial intervention and allows adaptation over time instead of delivering a finished standard product.

Light Renovation

A low-intensity intervention in which the layout, structure and building services are largely retained. Adaptation takes place through adjustment rather than extensive rebuilding.

Open Plan

A typology that retains large, continuous spaces for flexible use. Suitable where the existing structure already provides generous spans and ceiling heights, for example for cultural or commercial uses.

Culture and Production

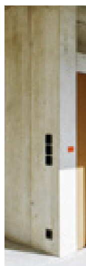
Flexible spaces for culture, production and public functions. Makes use of generous volumes and robust structures.

Shared Spaces

Shared kitchens, workspaces, workshops, laundry rooms, guest rooms or rental units can be grouped within the building, reducing the need for private floor area. This can activate dark, difficult-to-use or otherwise inaccessible parts of the structure and give them economic and social value.

Revenue

Cost / Climate Impact / Degree of Intervention



Full Service Living

Housing with integrated services such as cleaning, reception and shared facilities. Increases attractiveness and revenue potential by combining housing and services within the same structure.

Compact Living

Very small private units that depend on shared facilities outside the individual dwelling. Effective in buildings where façade conditions, daylight or structure limit the possibility of larger apartments.

Coliving

A housing model organised around private rooms and generous shared functions. It enables high density in deep office floor plans by centralising kitchens, living spaces and services.

Short-stay Living

Temporary accommodation, ranging from short stays to extended stay. Existing structures can often be used with relatively limited interventions.

Hotel

Standardised rooms organised around efficient logistics. Allows for clear operational structures but often requires adaptation of layouts and building services.

Senior / Intergenerational

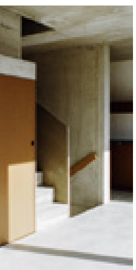
Organised around shared spaces and short distances rather than maximising private floor area. In repurposing projects, existing corridors and cores can become a social backbone rather than each apartment being optimised in isolation.

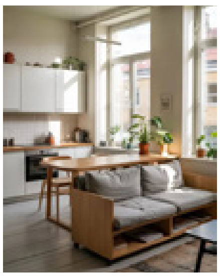
Student Housing

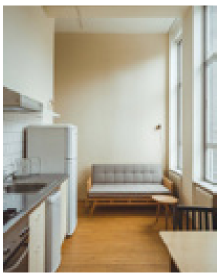
Space-efficient housing with an emphasis on shared facilities. Suitable for buildings where smaller units can be organised around communal spaces.

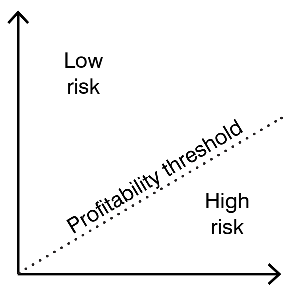
Standard Renovation

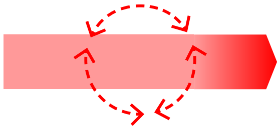
A standards-based upgrade in which layouts, building services and finishes are adapted to contemporary housing standards. Often requires more extensive demolition and gives less consideration to the existing spatial qualities.











Generative Scenario Assessment

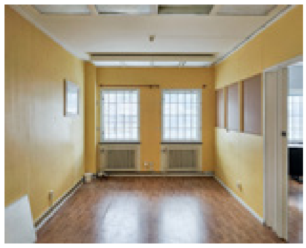
Generative Scenario Assessment is a method for understanding and comparing possible directions before a project is defined. It works as an addition to *Unlocking Existing Values*. An inventory of spaces, dimensions, structure, materials and technical conditions forms the starting point, bringing drawings, photographs and observations together.

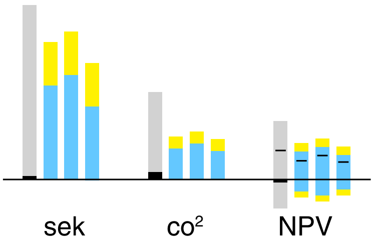
From this, one or several hypotheses are tested in parallel. Scenarios are generated with support from generative AI, using the existing building as the base and varying use, intervention and architectural direction. The generative part is not the answer, but a way of quickly opening and closing possibilities.

Each selected scenario is then translated into physical changes and assessed through intervention levels, cost and climate impact. Comparing the scenarios visually and through data makes it possible to understand their consequences without developing each one into a finished design.

The aim is to understand the **design universe that fits within the built framework**: what is possible, what is unlikely to work and where something unexpected may appear, before entering into details and the many hours of drawing that follow.

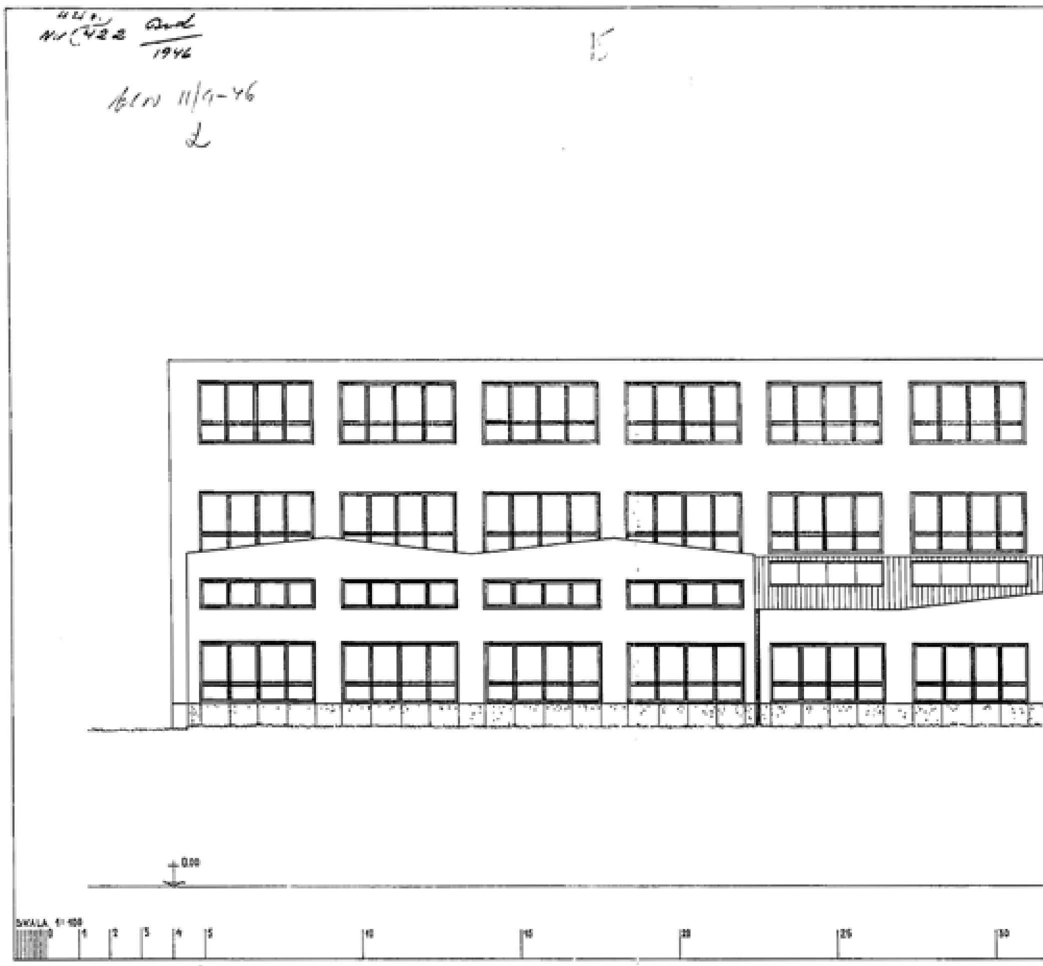
Could You Live Here?





Scenario ...

Instrumente



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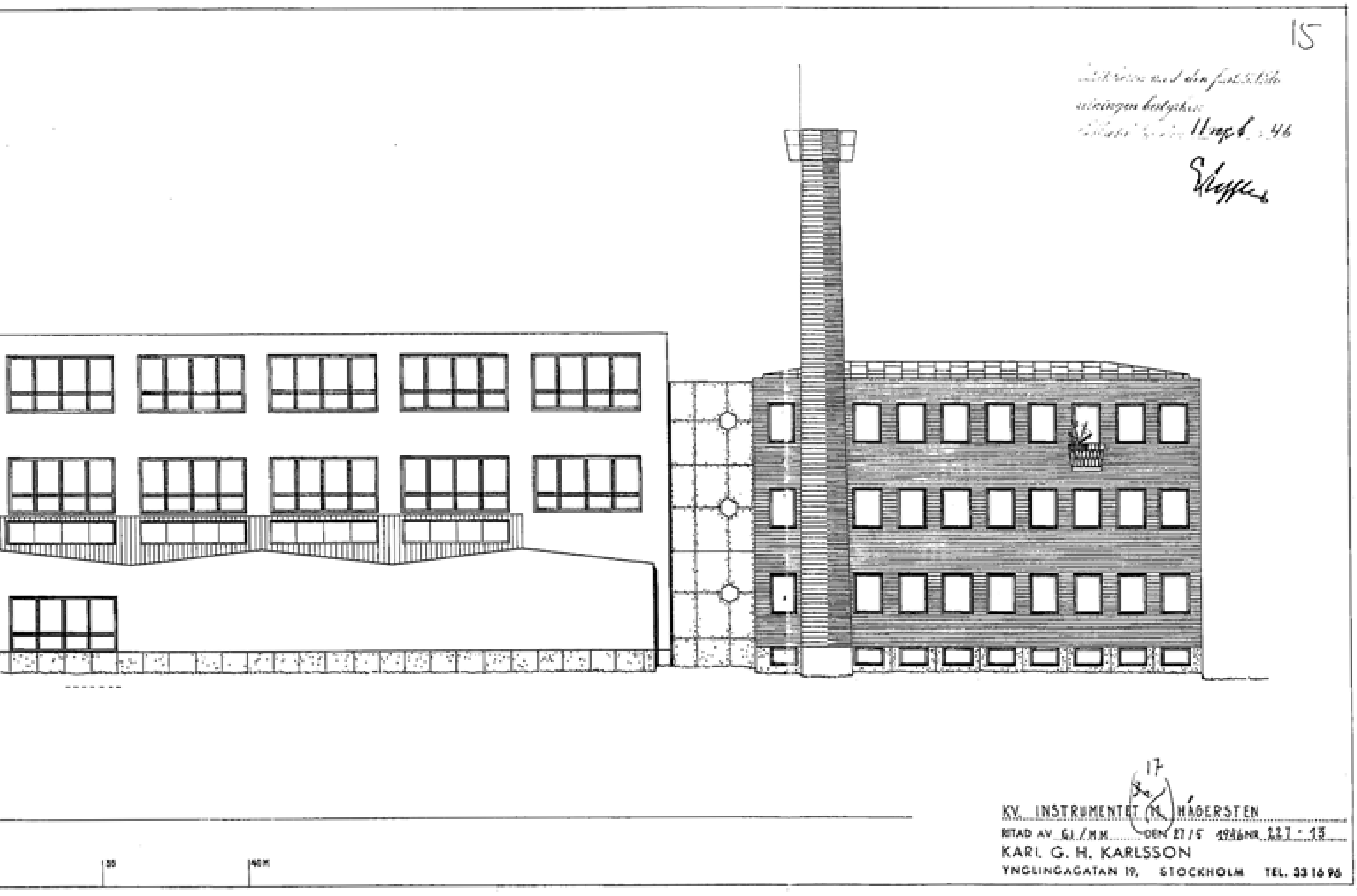

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17
KV. INSTRUMENTET HÄGERSTEN
RITAD AV GI./H.H. DEN 27/5 1946 NR. 227 - 13
KARL G. H. KARLSSON
YNGLINGAGATAN 19, STOCKHOLM TEL. 33 16 96

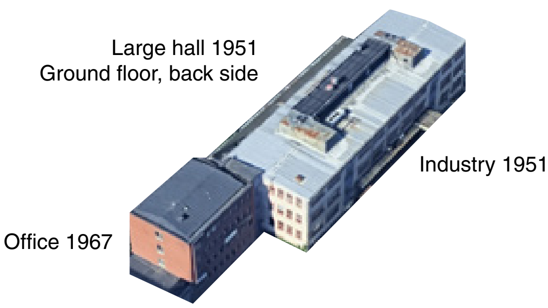
Örnsberg Stockholm

A long planning process has been running around how the industrial area of Örnsberg, Stockholm, could continue to develop and accommodate more housing. The detailed development plan was halted by politicians in 2023, asking: is it possible to preserve more of the existing buildings? Instead of demolishing almost everything, the process landed in identifying certain buildings with high cultural-historical value and the potential to be renovated.

This story still has a long way to go. The main difficulty is that developers and property owners have invested in the area based on calculations of being able to develop a certain amount of sellable or rentable square metres. Preserving buildings with less efficient footprints adds pressure on the rest of the area to be developed more densely. From the urban plan down to the concrete structure and its reinforcement, it is still not clear at the detailed architectural level whether the buildings planned to be preserved can accommodate the expectations placed on them. The buildings are standing, but they have to live up to a certain level, and this makeover cannot be too expensive either.

Together with the property owner Riksbyggen, I studied Instrumentet 17, testing and further developing the methods and tools Unlocking Existing Values and Generative Scenario Assessment. This resulted in a wider scope of typologies and possible renovation scenarios.

Örnsberg's industrial area was established mainly during the 1940s as the Jakobsdal workshop district, with small-scale industrial buildings for mechanics, metalwork and chemistry, sometimes combined with housing. The area was later supplemented with additional industrial and office buildings, while retaining its irregular structure and successive development. Today, its cultural-historical value lies largely in the whole: the small-scale industrial character, the varied building volumes and the legibility of its development over time. Being centrally located along the metro in southern Stockholm, the large, robust spaces and relatively low rents have also allowed a strong local community to take shape.


Large hall 1951
Ground floor, back side
Industry 1951
Office 1967

Inside and outside

1. The meeting between the industrial building, the office building and the street.
2. The office building has traditional cellular offices with two windows per room.
3. The industrial hall at the rear, at ground level, receives generous daylight from rooflights. Many layers of wear and alteration.

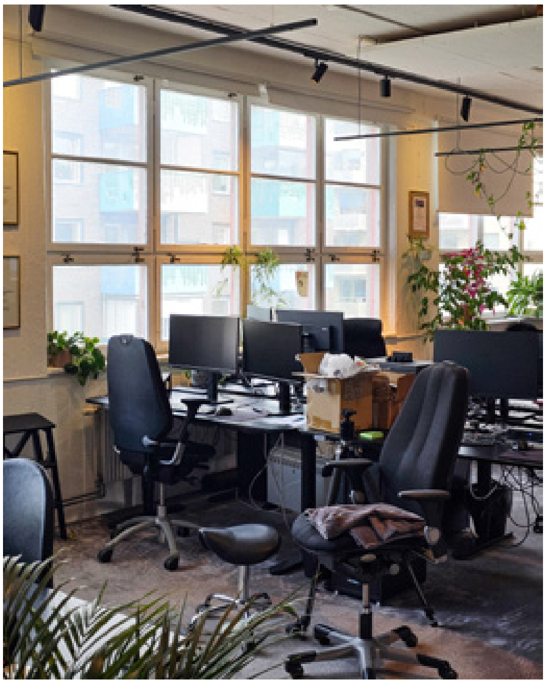
Industry and Office

The industrial building was constructed between 1946 and 1951 according to drawings by Karl G.H. Karlsson and has high cultural-historical value as part of the area's industrial core. The building is characterised by brick, blue render, a loading dock and a distinct industrial character, reinforced by the tall chimney as a landmark. The whole is considered sensitive, but can accommodate careful alterations and a restrained vertical extension.

The office building was constructed between 1961 and 1967 according to drawings by Allkopia Hans Fehnl and forms an important complement to the industrial building. It has façades of red brick, a clearly articulated stairwell in blue render, and a brick chimney that contributes to the coherent character of the area. The building is considered highly sensitive, but smaller additions towards the courtyard can be carried out with care.

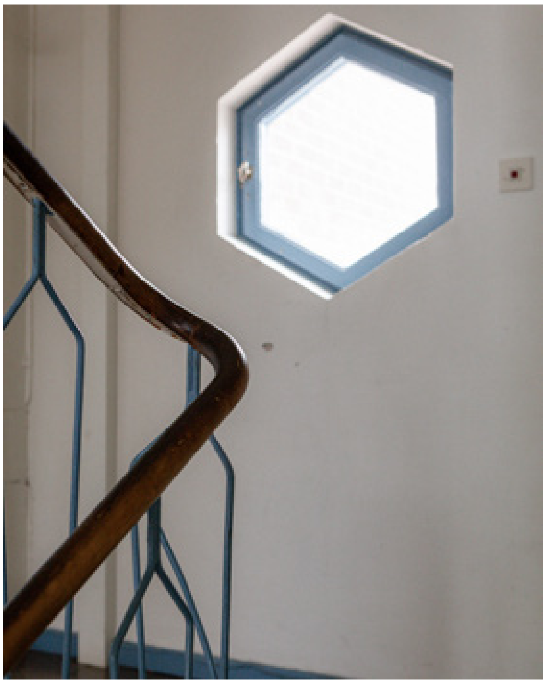
For Instrumentet 17, these conditions create a clear opportunity for development. The property is green-classified and has high cultural-historical values, while at the same time having a robust industrial structure that allows for adaptation. There is room to add, alter and reprogramme without losing the character of the whole. This creates a concrete scope for action in which new volumes and uses can be tested against what is already there.

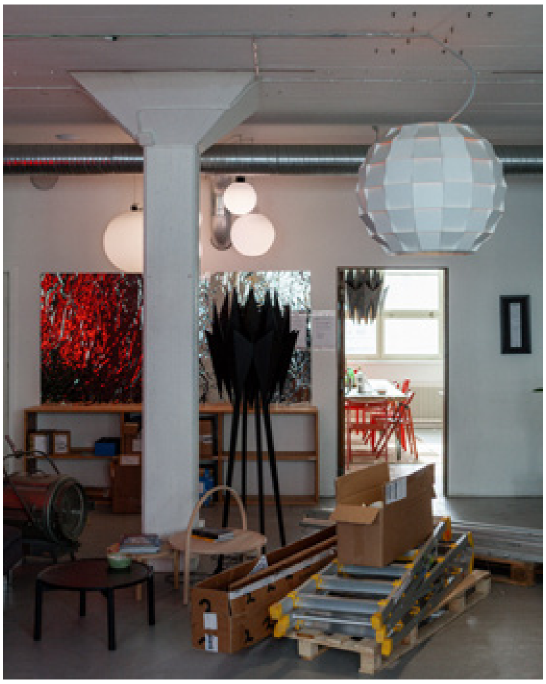
BTA total 4,900 sqm
Window area 595 sqm
Window to BTA ratio 12%
Floors 3–4 + b.



Current use

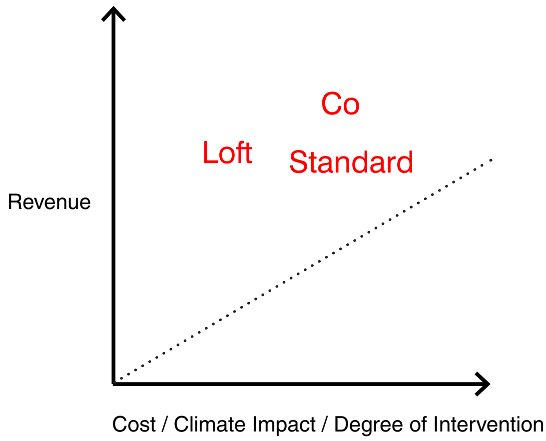
1. Office spaces on the top floor of the industrial building.
2. Workshop on the half-level, with mixed use and collective organisation.
3. Office spaces in the office building.





Space, structure and details

1. High ceilings, cast concrete, few installations and a clear functional expression.
2. Diamond-shaped window, timber handrail on metal balustrade.
3. Exposed cast-concrete column structure, high ceilings and visible services.



<u>Scenarios: Residential Conversion</u>

With the current direction of the urban plan for Örnsberg, Instrumentet 17 is to be preserved. However, to reach the same development potential as previously, the industrial building gains two additional levels in wood and a new multistorey apartment building is introduced at the rear. Building on this strategy of densification, drawn by Belatchew Architects, housing becomes a central question.

The existing structural frame, vertical circulation and façade rhythm are treated as starting conditions, while programme, subdivision and degree of intervention are allowed to vary. Through the initial mapping, three typologies emerged as particularly compatible with the building: loft living, standard apartments and coliving. They respond differently to the existing plan, structure and technical conditions, providing three distinct ways of testing how far the building can be adapted for housing.

The purpose at this stage is not to solve the apartment plan, but to establish where a residential programme fits the building and where the building begins to resist it. A rough planning study defines enough of the organisation, surfaces and circulation to move forward into generation and assessment without fixing a final design.

What changes from scenario to scenario is how much is asked of the existing building: whether walls are retained or removed, windows repaired or enlarged, how much new servicing is introduced, and how complete the residential fit-out becomes. The scenarios test different balances between adaptation and retention within the same physical framework.

Instrumentet 17

General input		**Description**
Project	Instrumentet 17	Instrumentet 17 lies in Örnsbergs industriområde in Hägersten. The property consists of connected industrial and office volumes plus a lower hall/garage volume.
Building year	1946 (industry) 1961–1967 (office)	
Adress	Jakobsdalsvägen 13-15 126 53 Hägersten	**Construction**
Value (SEK/m² BTA)	~10000	Office volume: four storeys with basement, concrete frame and slabs, red brick façade and blue plaster at the staircases. Industry volume: three levels, concrete structure and slabs, plaster façades, red-painted steel windows and pitched metal roof. A lower hall/garage extends partly under ground on
BTA total (m²)	4913	
Window area (m²)	595	**Cultural value**
Window ratio (to BTA)	12%	Green-classified with high cultural-historical value. Readable functionalist industrial building; the chimney is a strong landmark and identity carrier.
Floors (count)	4	
Wall to floor ratio	1.32	
		Location
		Between Örnsberg and Axelsberg. Strong accessibility and a clear link to Instrumentvägen and Jakobsdalsvägen.
Note		**Maintenance**
		Major maintenance backlog. Ventilation is largely not functioning. Heating still keeps the building running. Roof and windows have been patched over time and broader upgrades were deferred.

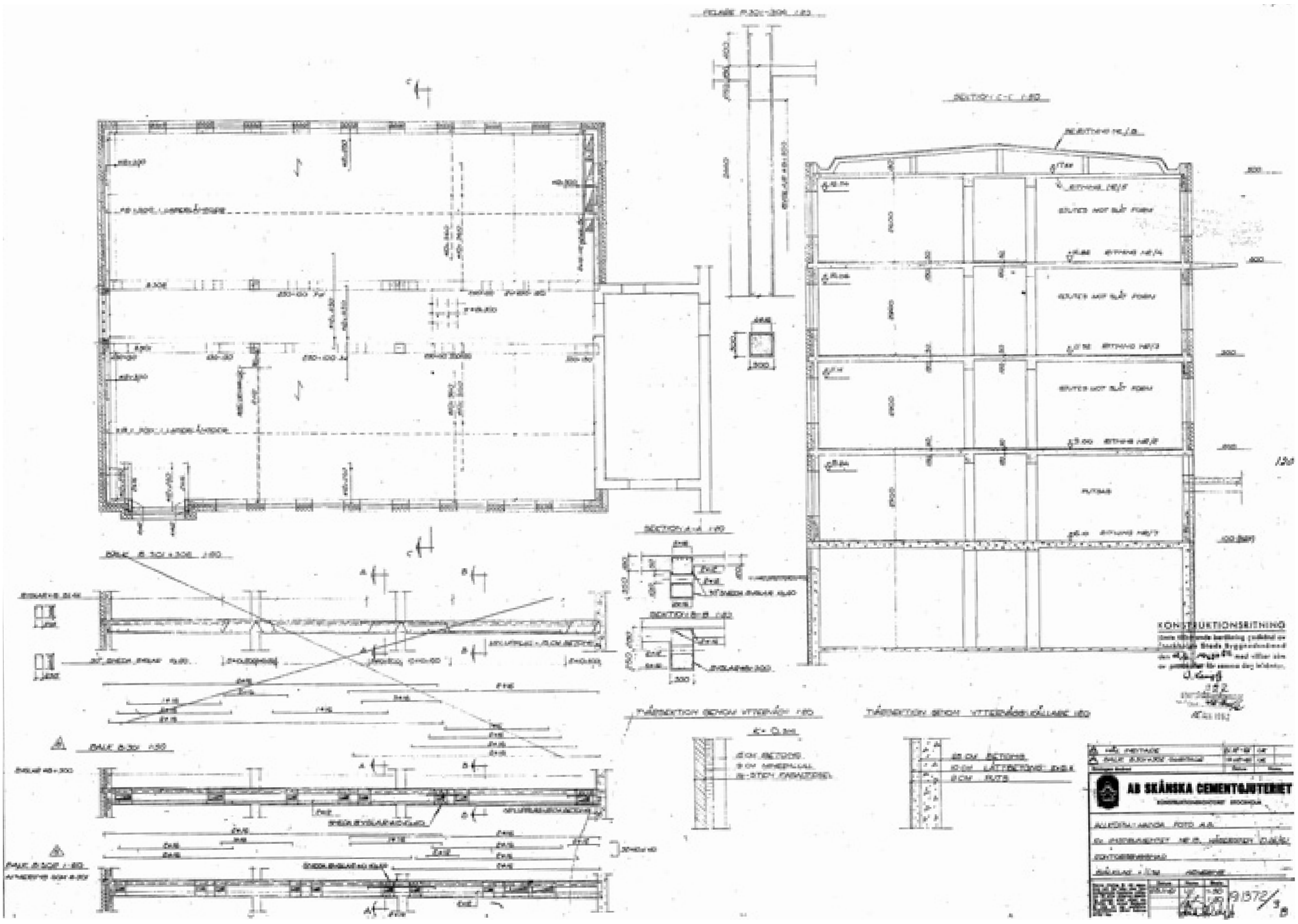
Existing

Level	BTA (m²)	BOA (m²)	LOAx (m²)	LOAy (m²)	Clng hght. (	Notes
Industry volume	**3220**	**0**	**1935**	**645**		
Plan 12	805	0	645		3.10	
Plan 11	805	0	645		3.10	
Plan 10	805	0	645		3.10	
Plan 09	805	0		645	2.70	
Office volume	**1050**	**0**	**660**	**165**		
Plan 13	210	0	165		2.60	
Plan 12	210	0	165		2.60	
Plan 11	210	0	165		2.60	
Plan 10	210	0	165		2.60	
Plan 09	210	0		165	3.00	
Hall volume	**643**	**0**	**611**	**0**		
Plan 10	643	0	611		4.00	
Project	**4913**	**0**	**3206**	**810**		
Efficency to BTA	*82%*	*0%*	*65%*	*80%*		
BTA upper floors (m²)	*2240*					
BTA ground floor (m²)	*1658*					
BTA below ground (m²)	*1015*					
Parking	*~23*					

Scenario 01 | Loft

General input		**Description**
LOAx	Office	Renovation of upper parts of industrial volume to loft living appartments. Office stays office. Ground floor recieves public.
LOAy	Other (lower rent)	
Time interval sustainability (y)	50	
Time interval finance (y)	30	
Ventilation system	level of intervention?	
Global etc	level of intervention?	
Global etc	level of intervention?	

Level	BTA (m²)	BOA (m²)	LOAx (m²)	LOAy (m²)	Clng hght. (	Notes
Industry volume	**3220**	**1290**	**0**	**1290**		
Plan 12	805	645	0		3.10	
Plan 11	805	645	0		3.10	
Plan 10	805	0	0	645	3.10	
Plan 09	805	0	0	645	2.70	
Office volume	**1050**	**0**	**660**	**165**		
Plan 13	210	0	165		2.60	
Plan 12	210	0	165		2.60	
Plan 11	210	0	165		2.60	
Plan 10	210	0	165		2.60	
Plan 09	210	0		165	3.00	
Hall volume	**643**	**0**	**0**	**611**		
Plan 10	643	0	0	611	4.00	
Addition	***1050***	***945***	***0***	***0***		
Plan 13	550	495			2.50	
Plan 14	500	450			2.50	
Project	**5963**	**2235**	**660**	**2066**		
Efficency to BTA	*83%*	*84%*	*79%*	*84%*		
BTA upper floors (m²)	*3290*					
BTA ground floor (m²)	*1658*					
BTA below ground (m²)	*1015*					
Parking	*~*					New bellow ground in block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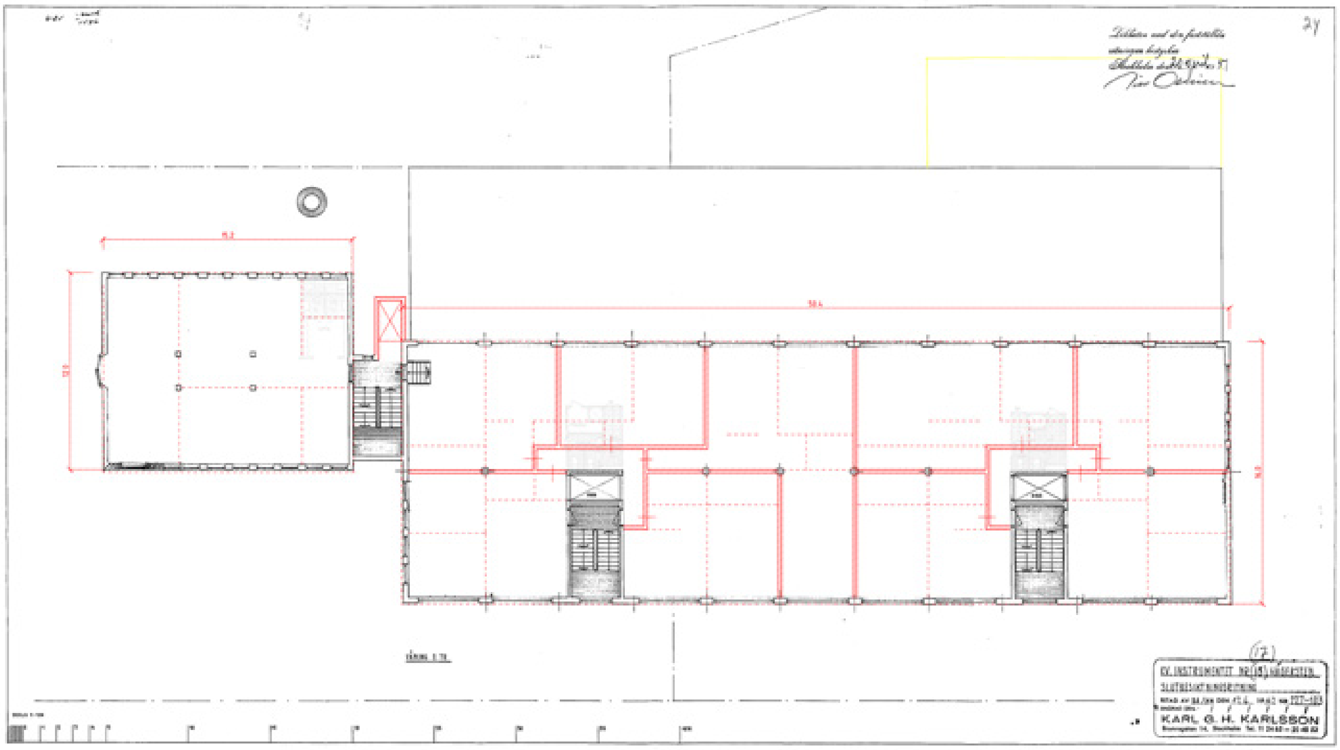


<u>Jumping Forward</u>

The plan is not solved, but through a first hypothesis its organisation gives an indication of surfaces, divisions and dimensions of spaces. With these basic parameters, the scenarios can be measured and assessed, allowing the process to jump forward and understand what is possible before getting stuck in details.

→ Loft Living

→ Standard Apartment

→ Coliving

→ s051 Chalet Chaleureux

The cellular office is opened into a generous loft where the existing columns and window rhythm give the home its character. New floors, windows and built-in joinery and wooden beams add warmth and order, while the larger rooms allow a relaxed way of living.

Cost	23,126 sek/sqm	■■■■□
Climate I.	239 kgco2e/sqm	■■■■■
NPV	107 Msek	■■■■□

s076 Plainly Raw

A restrained loft conversion that leaves traces of the former office visible. Selected walls are opened up, while the existing windows, radiators and structure remain. Cleaned concrete, simple finishes and a generous open room give it an informal, lived-in character. It’s yours to make.

Cost	15,637 sek/sqm	■■□□□
Climate I.	166 kgco2e/sqm	■■■□□
NPV	93 Msek	■■■□□

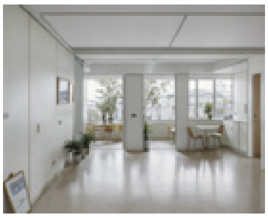
s196 Vertical Rhythm

A generous loft apartment shaped by the existing concrete frame and a more open plan. Enlarged windows and doors bring the city closer, while simple new floors and finishes keep the atmosphere calm, robust and slightly unfinished rather than overly polished. Cool, calm, simple.

Cost	20,645 sek/sqm	■■■□□
Climate I.	179 kgco2e/sqm	■■■■□
NPV	119 Msek	■■■■■

→ s340 Your Luxury

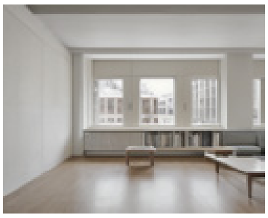
The apartment is organised around a bright living room opening to a new balcony. The original frame remains quietly present, while new windows, warm finishes and built-in storage make the former office feel settled and comfortable without erasing where it came from. Some simple luxury.



Cost	28,149 sek/sqm	■■■■■
Climate I.	233 kgco2e/sqm	■■■■■
NPV	82 Msek	■□□□□

s093 It’s Yours to Read

The former office becomes a generous apartment organised by a few carefully placed walls and built-in storage rather than a completely open plan. The existing facade rhythm remains, while renewed surfaces and joinery make the rooms practical, calm and distinctly domestic.



Cost	17,320 sek/sqm	■■■□□
Climate I.	156 kgco2e/sqm	■■■□□
NPV	99 Msek	■■■□□

s028 Brightness

A former office becomes a calm, generous apartment with wide views over Stockholm. The original structure and window rhythm remain legible, while new surfaces, windows and carefully integrated services make the space feel bright, quiet and distinctly residential.

Cost	20,458 sek/sqm	■■■□□
Climate I.	227 kgco2e/sqm	■■■■■
NPV	120 Msek	■■■■■

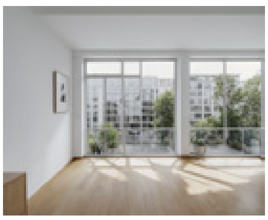
→ s084 Shared City Room

A bright shared home organised around a generous kitchen and dining room. The existing structural frame and window rhythm remain, while new herringbone floors, built-in storage and refined finishes make the common spaces warm, calm and domestic.

Cost	24,787 sek/sqm	■■■■□
Climate I.	219 kgco2e/sqm	■■■■■
NPV	99 Msek	■■■□□

s234 Fully Opened

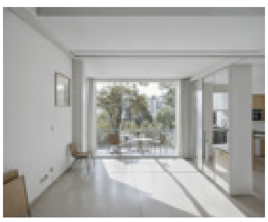
The former office opens toward the city through impressively large windows, while warm timber surfaces soften the retained concrete frame. Generous shared rooms feel distinctly domestic, with private spaces and new residential services fitted carefully around the existing structure.



Cost	26,438 sek/sqm	■■■■■
Climate I.	220 kgco2e/sqm	■■■■■
NPV	91 Msek	■■□□□

s339 Walk Outside

The shared room opens almost wall-to-wall toward the balcony, turning the city view into part of everyday life. The old concrete frame remains present, but new glazing, warm finishes and built-in elements make the interior feel generous, settled and distinctly domestic.



Cost	29,770 sek/sqm	■■■■■
Climate I.	244 kgco2e/sqm	■■■■■
NPV	74 Msek	■□□□□

Existing Space Instrumentet 17 Office

s028 Brightness Standard Apartment

Cost	20,485 sek/sqm	■■■□□
Climate impact	227 kgCO2e/sqm	■■■■■
NPV	120 Msek	■■■■■

<u>Evaluation: Residential Conversion</u>

The comparison changes the question from *can housing fit?* to *what kind of housing makes sense here?* Across the scenarios, the largest differences are not produced by architectural style, but by what has to be physically changed. Enlarged façade openings, new windows, structural alterations and complete technical upgrades quickly increase both construction cost and embodied climate impact. Retaining openings, structure and parts of the existing fabric has a much greater effect than whether the final interior appears raw or refined.

The scenarios also show that greater investment does not automatically produce greater long-term value. A more extensive conversion may support higher revenue, but this has to compensate for the additional construction cost. The interesting scenarios are therefore not simply those that do the least or the most, but those where the new use and the existing building are well matched.

For Instrumentet 17, loft and raw-living typologies become particularly relevant because they can accept qualities that standard housing would normally try to correct: generous rooms, visible structure, traces of previous use and a less complete fit-out. What is retained can become part of the value of the dwelling rather than simply a constraint.

s196 Vertical Rhythm

Cost	20,645sek/sqm	■■■□□
Climate impact	179 kgco2e/sqm	■■■■□
NPV	119 Msek	■■■■■

s339 Walk Outside

Cost	29,770 sek/sqm	■■■■■
Climate impact	244 kgco2e/sqm	■■■■■
NPV	74 Msek	■□□□□

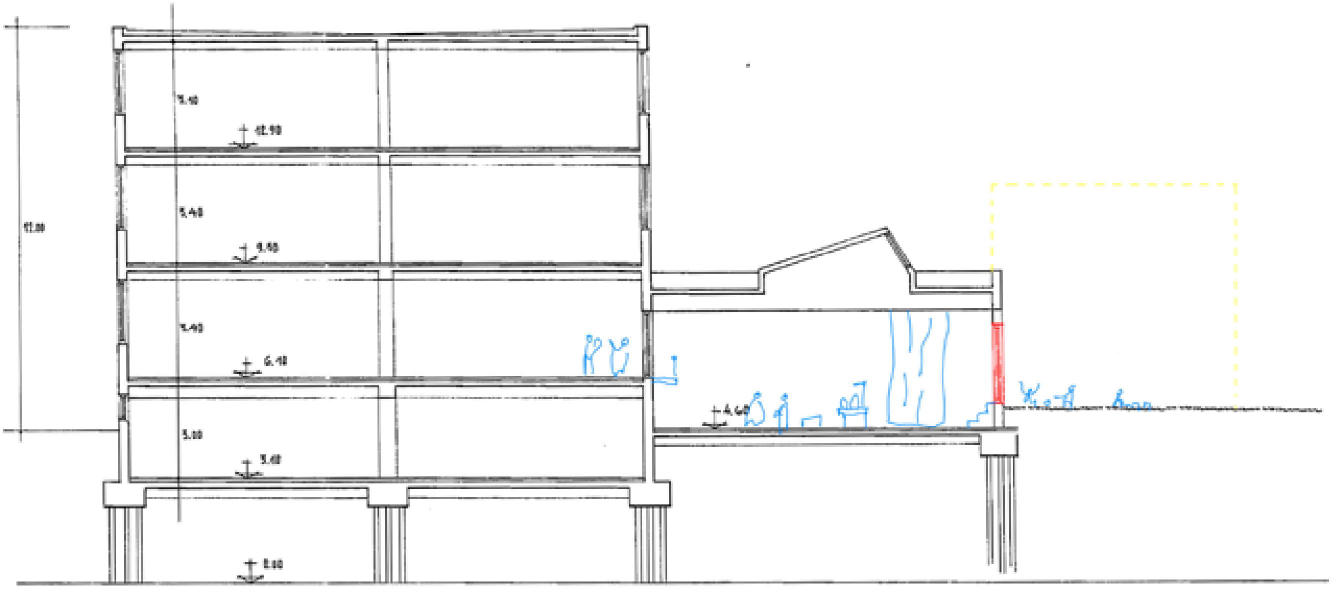
The office has been standing empty for a couple of years, and it is starting to show. Humidity, objects left behind, a space that no one takes care of. In the middle of the room we placed a table and a couple of chairs. What could be more fitting than talking about the future of this space within the space itself? We show spreadsheets, talk about the market, about what is happening in the area and in the world. We also talk about what the space could be, and we visualise it. Could this be its future, or this? It points in many directions, but sitting there I feel that something can actually be done. There is more here than a space left behind.

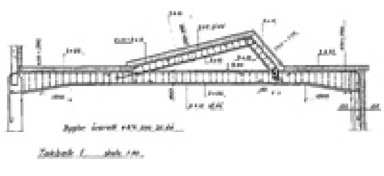
Scenario: Industry Hall

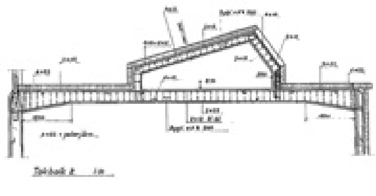
The hall introduces a different question. Its value lies less in how efficiently it can be subdivided than in the fact that a continuous room of approximately 490 sqm already exists. Rather than translating it into housing, the hypothesis is to preserve this spatial exception and make it usable through limited but targeted intervention.

The permanent partition through the hall is removed, original openings towards the courtyard are reopened, and the roof lights, concrete structure and overall volume are retained. Permanent additions are kept deliberately sparse. Temporary partitions and movable elements can instead support workshops, events, production, a café or other collective uses without fixing the room to one programme.

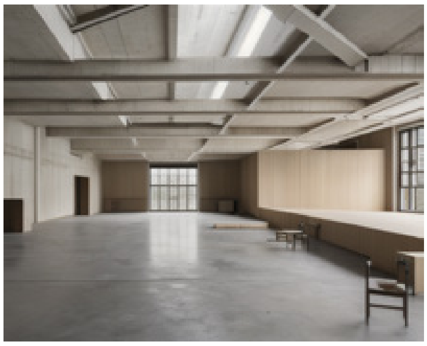
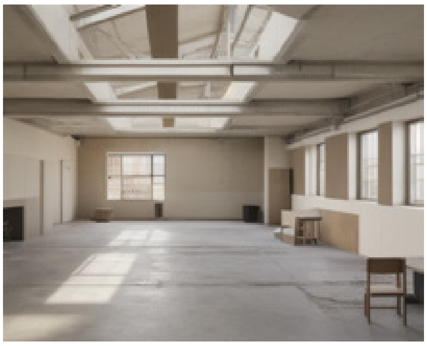
Here, uncertainty about use does not necessarily have to be solved before work begins. An activation phase can test who needs the space and how it is occupied, allowing the programme to develop through use. The scenario is therefore also temporal: repair and activate first, learn from occupation, and make further investment when there is a reason to do so.

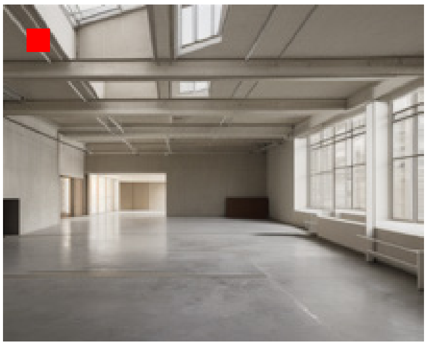
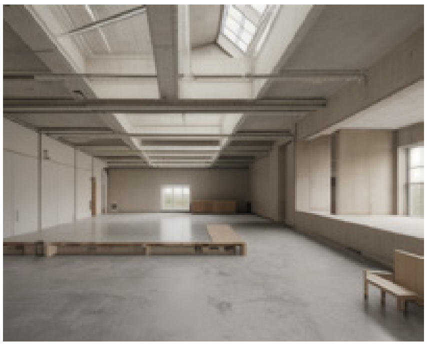
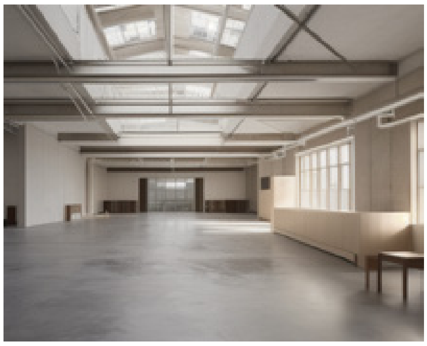








Existing Space Instrumentet 17 Hall

s011 Loft Living
Cost 13,442 sek/sqm
Climate impact 97 kgco2e/sqm

Evaluation: Industry Hall

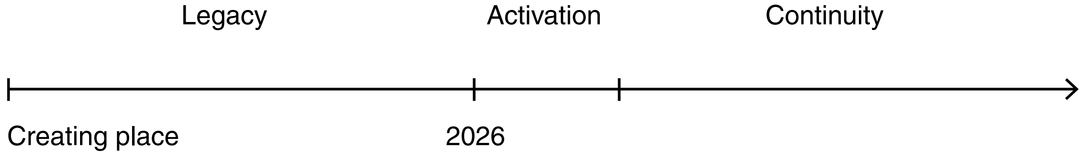
The hall scenarios show a different relationship between value and intervention than the residential conversion. Because the main volume, structure and spatial organisation are retained, both construction cost and embodied climate impact can remain substantially lower. The decisive interventions are primarily technical rather than spatial: ventilation, thermal performance, windows, lighting and electrical capacity, together with the requirements needed for public use.

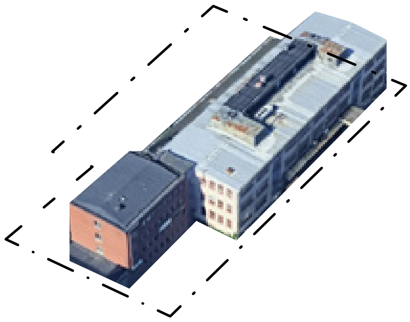
This shifts the role of design. The question is not how much architecture needs to be added to complete the room, but what can remain unresolved. Each additional layer of subdivision, finishing and technical specification makes the hall more specific and more expensive, while potentially reducing some of the openness that gives it value in the first place.

A staged approach therefore becomes part of the proposition. The first investment can make the hall usable without determining its final programme. Later interventions can respond to actual users and demands. In this case, doing less is not simply a conservation strategy; it is a way of preserving future possibilities.

s011 Loft Living
Cost 8,052 sek/sqm
Climate impact 50 kgco2e/sqm

s011 Loft Living
Cost 10,523 sek/sqm
Climate impact 71 kgco2e/sqm





Existing

Pinterest, Hemnet* or Boverket**?

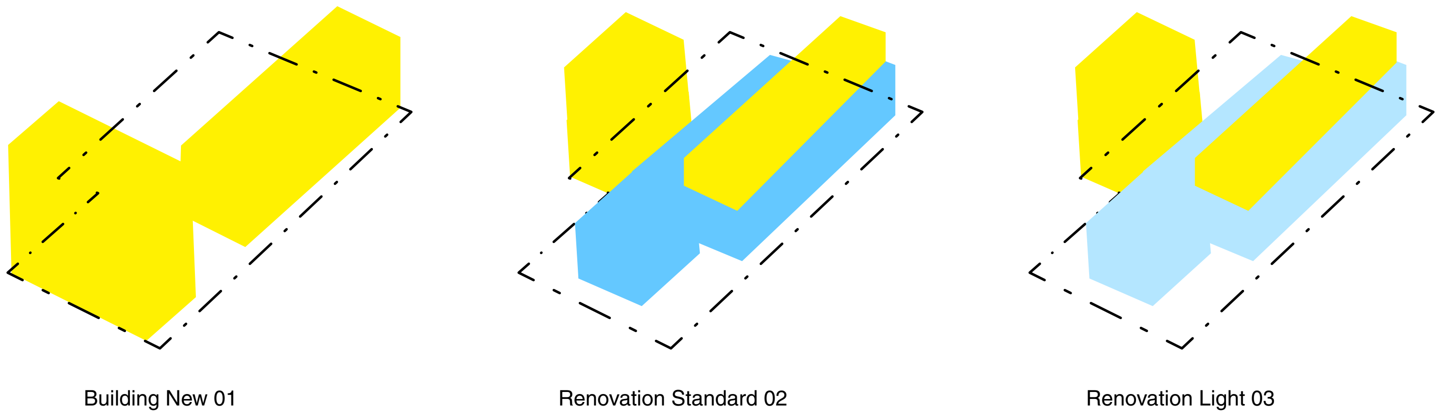
To start from the existing is to start from the local and the specific, in a place, a building, a person. It is to understand one's own time, history and oneself. What is our vision of housing and the good city? The desire to work more carefully with what is already there has shifted significantly in recent years, from a smaller sphere to something discussed across architecture, planning, politics and the property sector. Another view, and another will for how the city should develop, is taking shape.

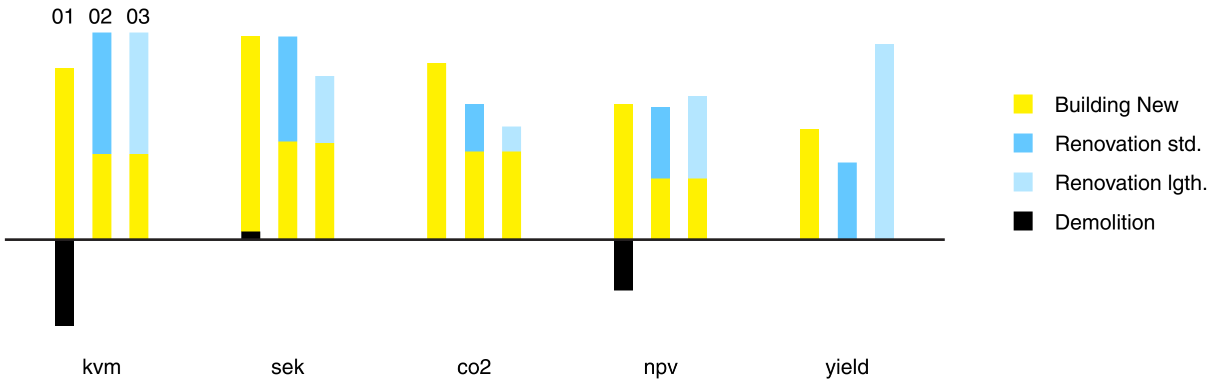
Following several properties over these years has made one thing clear: the question is rarely simply whether a building can be kept. It is what we ask it to become, under which rules, and with which expectations of value. Instrumentet 17 can be understood as inefficient office and industrial space, as housing, as a collection of robust rooms, or as an industrial hall whose value lies precisely in not being subdivided. The answer changes when the question changes.

How a property is developed stands or falls on whether the economy can be made to add up, from its smallest parts to its whole in the city. This is a reality that will not change. What can change are the questions and frameworks we bring to it. What do we want, before we answer what we can? The process is not linear, and neither has this investigation been. New knowledge about structure, regulation, economy or use can redraw the direction of a project. One discovery leads to another, trying to understand not only what should be done, but how we do this at all.

This movement is also happening around the projects. Research, politics and regulation are changing, in Sweden and elsewhere. Boverket has in recent years investigated how conversion can be made easier, while the Swedish government's 2026 proposal for changes to the Planning and Building Act would limit many requirements at alteration to the part of the building actually being changed, rather than allowing the intervention to trigger extensive requirements for the whole building. If adopted, such changes alter not only what is permitted, but what becomes reasonable to imagine in the first place.

The image is useful within this process because it gives a discussion something to gather around. Can this be done within the framework of the existing, could you live here?. Where the next question often is would you buy it? Continuously, what is seldom discussed, is it also possible to shift the economic framework of the existing and ask, who do we want to live here?





Assessment of Instrumentet 17 as a whole: the existing building is renovated, two timber floors are added and a new volume is placed in the courtyard, compared with the previous proposal to build entirely new.

This is where Pinterest, Hemnet and Boverket meet. Pinterest is desire, atmosphere and imagination. Hemnet is the market, saleability and domestic aspiration. Boverket is regulation, construction and standards. Architecture moves somewhere between all three, having to touch all these points, but also lifting the discussion to not just end in the image of the beautiful home and the number of the price.

The generative AI image, what is sometimes referred to as synthetic architecture, sits strangely within this. It plainly does not exist. It has no budget, structure, fire strategy or boundaries beyond those we give it. This is almost the opposite of the existing environment, where the boundaries are literal and often what makes the project interesting. They are uncommon friends. One expands the universe of imagination, while the other constantly narrows it through what is already there. Combining them can be productive: the image pushes outward, the building answers back. Somewhere between the two, possibilities appear.

It should also be said that visualisation is a misleading tool. We have seen far too much architecture floating effortlessly in images, only to land rather crookedly when built. Generative images make this even easier. Their value here is not as representations of a finished project, but as hypotheses that can be challenged. Once the image is brought back to structure, intervention, cost, climate impact and regulation, it becomes possible to ask not only whether we like it, but what it demands and what it gives back.

The conclusion can never be that every building should be retained. Before demolition is accepted as the rational outcome, what is already there should be given a real chance to show what it can do.

The larger change in how we build will take shape through specific cases: individual buildings, particular conditions and the people making decisions around them. These are the stories through which our processes, regulations and visions gradually change.

There are people behind. There are buildings too.

* Hemnet is Sweden's main platform for buying and selling apartments and houses, a marketplace that also gives a view into the desires surrounding our home environment.

** Boverket is Sweden's National Board of Housing, Building and Planning, responsible for building regulations as well as investigations and initiatives concerning the built environment. Its work is expert-led, but the final framework is political and ultimately democratic.

Latent Potential

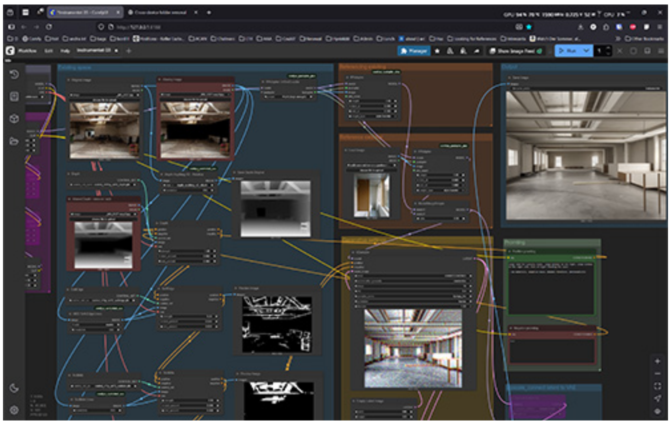
Latent space can be understood as an intermediate space where spatial qualities, materiality and atmosphere are represented as relationships rather than finished objects. In image diffusion, an existing photograph is translated into an abstract representation of light, proportion, texture and spatial logic, from which an image is gradually reconstructed.

Within this space, several interpretations of the same starting point can coexist. Small shifts can produce distinctly different directions: a room may become residential rather than office, raw rather than refined, open or more defined. The images are therefore not variations of a finished proposal, but parallel scenarios rooted in the same existing space.

The visual process is deliberately tied to the building. Photographs form part of an inventory of its structure, plan, technical systems and previous use. Geometry and proportions are extracted from the existing image and used to retain spatial relationships while new layers are introduced. Materiality, use and degree of intervention can then be varied without losing the underlying logic of the room.

This becomes an analogy for the latent potential of the building itself. Within the existing structure are relationships between light, dimensions, structure and material that may not yet be fully articulated. By working in this intermediate space, before a direction is fixed, different futures can be tested without first changing the physical building.

The result is not a prediction of what the room should become, but a spectrum of possible development paths. Latent potential becomes a way of making what is not yet realised visible and comparable. Uncertainty is not eliminated, but made readable.



From Image to Assessment

The evaluation is an automated workflow moving between image generation, language analysis and calculation. Each generated scenario is compared with the existing condition and translated into a structured description of what would physically have to change. The image is not calculated directly; it becomes a set of assumptions that can be tested against the building.

Changes are divided into building elements such as demolition, structure, windows, façade, internal walls, floors, kitchens, bathrooms and technical systems. Each is assigned an intervention level from 0–3: no intervention, light adaptation, substantial alteration or comprehensive replacement. A raw-looking interior can therefore still contain expensive structural work, while a more finished scenario may retain most of the existing building.

Construction Cost (sek/sqm BTA)
Each intervention is connected to a quantity, an affected share and a unit cost. Property-specific quantities, such as 4,913 m² BTA and 595 m² of windows for Instrumentet 17, form the base of the calculation. The work items are summed and supplemented with allowances for establishment, design and consultants, project management and early-stage uncertainty. The result is an early comparative construction cost rather than a tender estimate.

Climate Impact (kgco2e/sqm BTA)
The same intervention structure is used for climate impact. Retaining an element adds little or no new embodied impact, while replacement and new construction introduce new materials and emissions. The calculation therefore makes the degree of reuse directly visible in relation to the proposed architectural change.

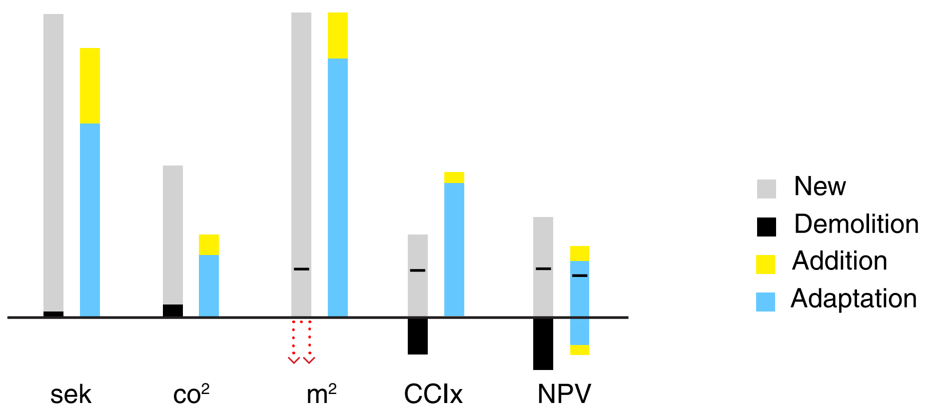
Net Present Value (NPV)
The construction estimate is then connected to the property model. For Instrumentet 17, NPV is calculated over 30 years with a 3% discount rate. Different scenario levels also adjust the assumed rental income, while investment changes according to the calculated renovation cost. A more extensive scenario can therefore create greater revenue, but only in relation to the additional investment required.

The purpose is not to predict an exact future cost or value. It is to run different architectural scenarios through the same chain of assumptions, making their consequences comparable in terms of intervention, investment, climate impact and long-term value.

	A	B	F	G	O	P	Q	R	S	T	U	V	W	Z
1	Index	g/s	Translation	In.	Translation	Cost datapoint	unit	EUR/unit	SEK/unit	Amount	Affected element/ Amount	SEK/m²BTA	Detailed cost SEK/m²BTA (used)	A1-A3 kgCO2e/unit
2	10-1	s	Remediation & light demolition	1	Selective demolition, retaining suspended ceilings/installations where possible; removal and sorting of small amounts of construction waste.	BKI-384.001.180-11.25 Gypsum partition wall demolition and disposal	m²	€59.30	611 kr	736.95	0.30	27.51 kr	25.16 kr	
3						BKI-384.001.200-11.25 Floor tiles removal and disposal	m²	€28.11	290 kr	60.00	0.20	0.71 kr		
4						BKI-384.001.204-11.25 Wall tiles removal and disposal	m²	€25.64	264 kr	300.00	0.20	3.23 kr		
5	10-2	s	Remediation & light demolition	2	Partial demolition into larger parts (floors, ceilings, some walls); sorting and container logistics.	BKI-384.001.231-11.25 Suspended ceiling demolition and disposal	m²	€24.91	257 kr	1,050.00	0.50	27.43 kr	94.59 kr	
6						BKI-384.001.180-11.25 Gypsum partition wall demolition and disposal	m²	€59.30	611 kr	736.95	0.50	45.85 kr		
7						BKI-384.001.200-11.25 Floor tiles removal and disposal	m²	€28.11	290 kr	60.00	0.40	1.42 kr		
8						BKI-384.001.204-11.25 Wall tiles removal and disposal	m²	€25.64	264 kr	300.00	0.70	11.29 kr		
9	10-3	s	Remediation & light demolition	3	Total tearing down to the frame in most surfaces (including older installation parts); higher requirements for protection, dust management and a...	BKI-384.001.231-11.25 Suspended ceiling demolition and disposal	m²	€24.91	257 kr	1,050.00	0.80	43.89 kr	570.00 kr	
10						BKI-384.001.180-11.25 Gypsum partition wall demolition and disposal	m²	€59.32	611 kr	736.95	0.80	73.36 kr		
11						BKI-384.001.182-11.25 Masonry interior wall (11.5 cm) demolition and disposal	m²	€50.54	521 kr	491.30	0.80	41.67 kr		
12						BKI-384.001.200-11.25 Floor tiles removal and disposal	m²	€28.11	290 kr	60.00	0.80	2.83 kr		
13						BKI-384.001.204-11.25 Wall tiles removal and disposal	m²	€25.64	264 kr	300.00	0.80	12.51 kr		
14						BKI-387.000.012-11.25 Sorting and disposal of potentially dangerous construction waste	t	€268.67	2,769 kr	24.57	1.00	13.84 kr		
15						BKI-387.000.009-11.25 Sorting and disposal of non-dangerous construction waste	t	€425.00	4,380 kr	393.04	1.00	350.29 kr		
16	20-1	g	Hole making / shaft	1	Small holes in plaster/lighter construction for electrical/plumbing; limited restoration.	BKI-313.000.179-11.25 Drilling installation lines and closing them with mortar after installation	m	€89.48	922 kr	14,739.00	0.01	27.66 kr	22.13 kr	
17	20-2	g	Hole making / shaft	2	Several holes and penetrations for new shafts/ducts; some drilling in concrete/brick; restoration of surface layers locally.	BKI-313.000.179-11.25 Drilling installation lines and closing them with mortar after installation	m	€89.48	922 kr	14,739.00	0.03	82.59 kr	91.85 kr	
18						BKI-313.000.186-11.25 Drilling shafts through concrete ceiling, up to 50x20 cm and 20 cm thickness	piece	€81.17	837 kr	12.00	0.25	0.51 kr		
19	20-3	g	Hole making / shaft	3	Extensive drilling through concrete/flooring for new vertical shafts and fire sealing; more extensive restoration.	BKI-313.000.179-11.25 Drilling installation lines and closing them with mortar after installation	m	€89.48	922 kr	14,739.00	0.06	165.99 kr	210.37 kr	
20						BKI-313.000.186-11.25 Drilling shafts through concrete ceiling, up to 50x20 cm and 20 cm thickness	piece	€81.17	837 kr	12.00	0.50	1.02 kr		
21						BKI-313.000.199-11.25 Cutting reinforced concrete, up to 43 cm thickness	m	€262.86	2,709 kr	98.26	0.10	5.42 kr		
22	21-1	g	Frame – joists & beams	1	The frame is retained; only minor repairs, fastenings and inspection/remediation of local damage.	BKI-312.000.249-11.25 Skilled worker hour	h	€93.80	967 kr	393.04	0.08	5.80 kr	4.64 kr	
23	21-2	g	Frame – joists & beams	2	Local reinforcements (e.g. in new shafts/openings), supplementary load-bearing structures and fire/noise measures linked to re...	BKI-312.000.249-11.25 Skilled worker hour	h	€93.80	967 kr	393.04	0.38	29.61 kr	38.45 kr	
24						BKI-312.000.189-11.25 Door openings in masonry, up to 30 cm thickness, including disposal of material and new lintels	m²	€189.90	1,957 kr	24.57	0.40	3.51 kr		8.7 kg
25						BKI-312.000.186-11.25 Window openings in masonry, up to 40 cm thickness, including disposal of material and new lintels	m²	€197.00	2,030 kr	12.28	0.40	2.03 kr		19.3 kg
26	21-3	g	Frame – joists & beams	3	Extensive structural reconstruction with major interventions (new openings, cast-in-place, major reinforcements).	BKI-312.000.249-11.25 Skilled worker hour	h	€93.80	967 kr	393.04	1.00	77.36 kr	460.18 kr	
27						BKI-312.000.189-11.25 Door openings in masonry, up to 30 cm thickness, including disposal of material and new lintels	m²	€189.90	1,957 kr	24.57	1.00	9.79 kr		8.7 kg
28						BKI-312.000.186-11.25 Window openings in masonry, up to 40 cm thickness, including disposal of material and new lintels	m²	€197.00	2,030 kr	12.28	1.00	5.08 kr		19.3 kg

Earlier Study: Primus 1

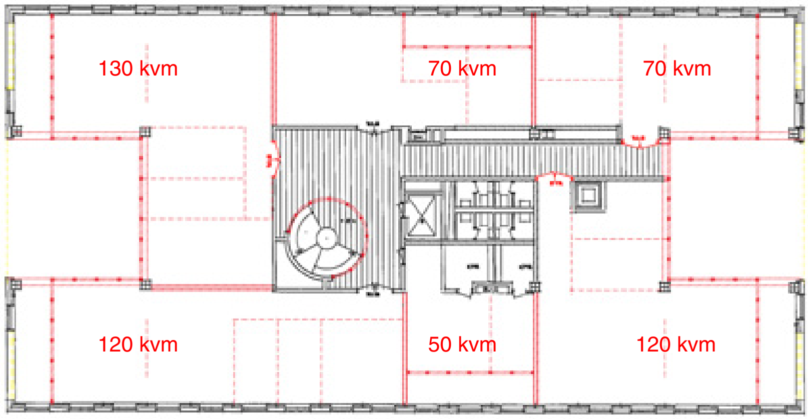
In opening up the deep plans, the first floor is accommodating flexible workspaces, then apartments, topped by a lightweight addition in wood. Left as unwritten sheets, the residents can themselves imagine their homes, a loft living typology. Two lightwells are opened in the structure and outdoor loggias added. Costs are kept down by frugal and precise interventions. Primus 1 became the case where *Could You Live Here?* took shape: was the building the problem, or had its potential simply not been explored?



Construction Cost (sek) - Building costs (average value, 4'000 €/ m²) are compared with the renovation alternative of the loft living apartments. Addition in wood of higher standard. Early surveys report no hazardous contamination in the structure
Life Cycle Analysis 50y. (co²) - Climate impact (average value, 310 kgCO_2e/m²) of multistorey apartment blocks is compared to an alternative where the energy consumption is slightly higher, but with substantially lower construction cost due to the reuse of the existing structure. Garage and foundation kept in both scenarios. Simplified LCA over 50 years. Covering A1–A5, B6, C1–C4; adding module D (circular benefits) is a next step.
Total amount of square meters (m²) - Both scenarios host the same number of square metres, though in the case of a new construction, a slightly higher amount can be rented out / sold.

Cultural Comparative Index (CCIx) - A speculative measurement that can only be used in comparison between different scenarios. Mapping values in the present, what benefits could arise from new changes and what could be lost in this process. Based on methods of cultural heritage and studies of social values.
Net Present Value 50y. (NPV) - The expected value of the property in 50 years measured in today's terms. Different from the construction cost, though related. New construction with a higher level of investment will also yield a greater risk. The renovation project is calculated to have a lower income and a higher maintenance cost. In extension, the Benefit-Cost Ratio (BCR) can show how a lower rate of investment could be benificial.
Net Value (—) - The sum of the positive and negative values of the measurement.

Typical plan:
680 sqm BTA
550 sqm BOA
80 sqm Loggia
115 sqm Dmlsh
0.81 < BOA/BTA

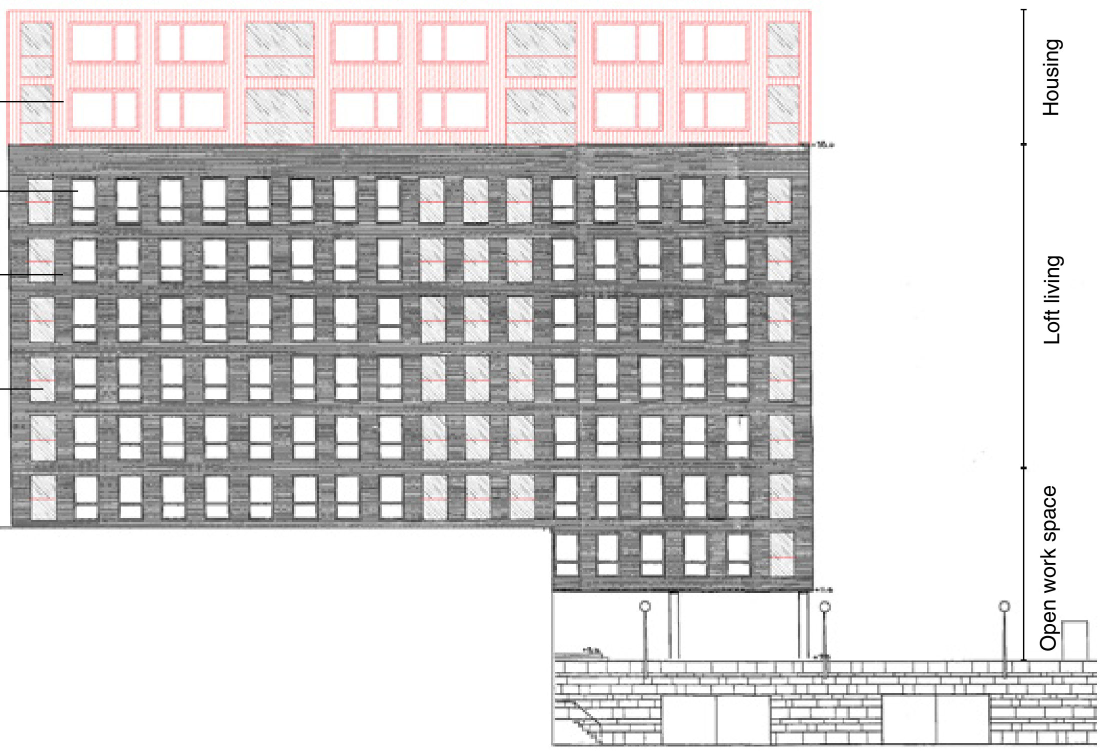


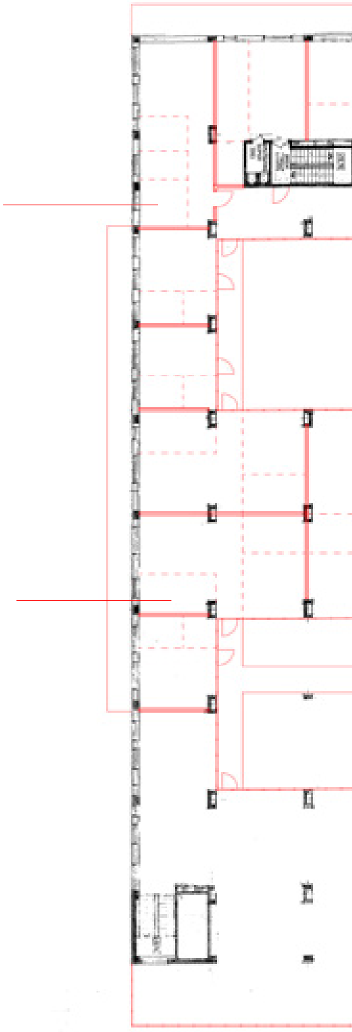
Light wood addition

Active facade natural vent.

Renovating existing facade

Loggia

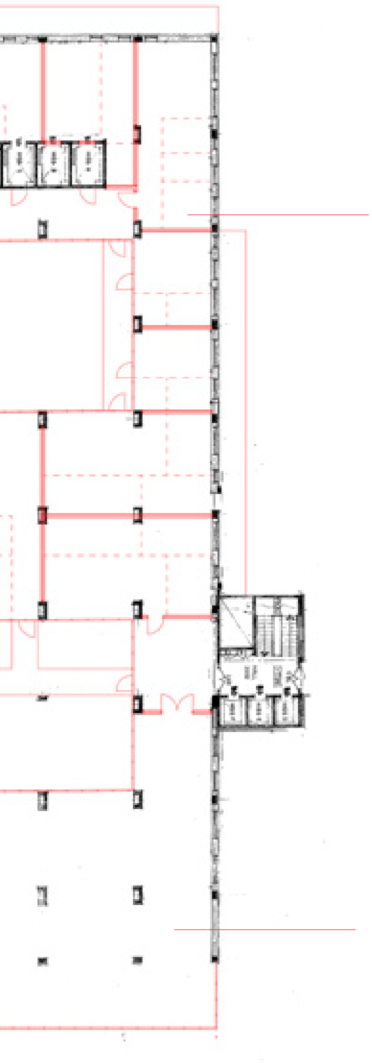


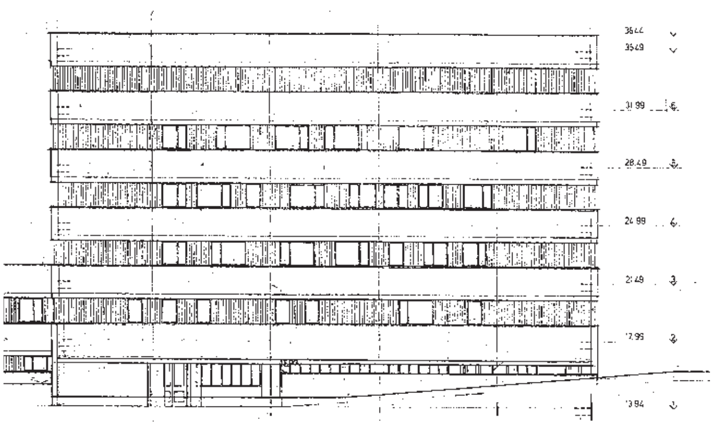


Earlier Study: Tegelbruket 4

Tegelbruket 4 has a deep but rational structure, built around a column-and-slab system with a non-load-bearing façade. A conversion hypothesis opens two courtyards within the existing frame, bringing daylight into the depth of the building and allowing offices and housing to coexist. Towards noisy Fleminggatan, offices remain; elsewhere, more open and generous apartments are introduced. The question was less whether the building could be reused, but how that reuse could fit within the larger urban plan and its vision of the traditional city.

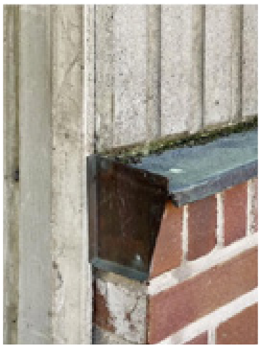
The study became an early test of the process later developed as Generative Scenario Assessment. A rough conversion hypothesis was used to jump forward, test what the structure could accommodate and understand which questions needed to be answered before going into a full design.





Inventory & Mapping

Site visits, documentation, surveys, use, flows, physical condition, social values, cultural values, material inventories and existing systems.



Scenario Exploration

Testing expressions, spatial configurations, materials and uses through rapid AI visions. An expanded basis for identifying the possibilities within an existing building.

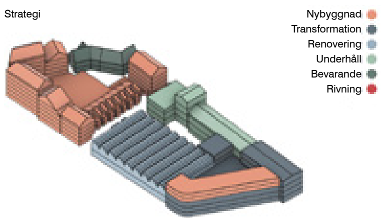
↓

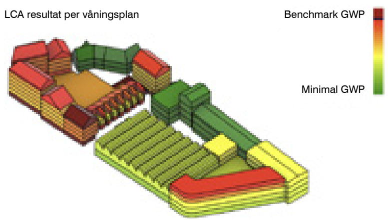
Structure & Constraints

Understanding the building's parameters. Structure, categorisation, load-bearing systems, dimensions, grids, floor heights, daylight, cores, services, constraints and opportunities.

LCA + LCC volumes

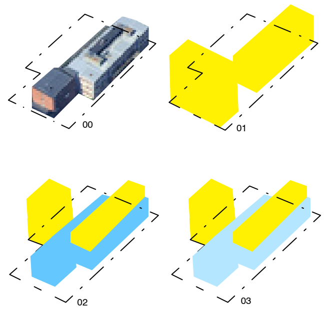
Early volume studies compare scenarios, linking quantities to indicative cost, climate impact and risk. Estimates are based on statistical ranges, benchmark data and different strategies for new construction, renovation, maintenance, preservation and demolition. Developed by Aarhus University in collaboration with Cebra.

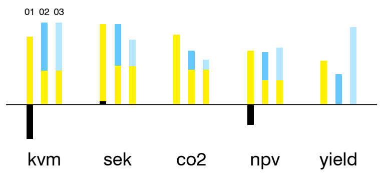




Scenario Evaluation

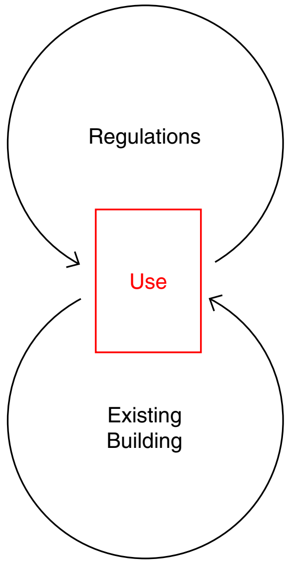
Alternatives, comparisons, risks, uncertainties, feasibility, cost, climate impact, values and priorities.





Matching Typologies

A continuous dialogue between the building's physical conditions and the requirements that may apply. Exploratory work in relation to changing regulations, combined with early dialogue with authorities to identify the best approach for the future development of the property.



Expanding Methods and Tools

The process is deliberately mixed. Drawings, surveys, images, calculations and regulations are used side by side, each answering different questions and correcting the others.

End Notes

Selected sources:

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This work and publication could not have been done without the generous financial support of Ivar & Anders Tengboms fond, Riksbyggens Jubileumsfond: den goda staden and Helge Ax:son Johnsons stiftelse. For the exhibition, I extend my thank you to the support from Stiftelsen Längmanska kulturfonden.

Not to mention the support from my fiancé Martina Hopfner, family, friends and colleges who have been part of enriching the discussion.

This publication was launched on the 4th of September 2026 for the exhibition of the same title at SPARK Gallery in Malmö.

Physical model of Instrumentet 17 made with Zahra Zeyad Aldasoki and Hugo Oskarsson.

Generative Scenario Assessment (GSA) calculations are made in collaboration with Markus Gustafsson (MSc Architecture & MSc Engineering, Chalmers) and Tomislav Martinovic (MSc Architectural Engineering, MEng Construction Management and Technology).

Font Helvetica and Consolas.

Page 14-15, references of AMUNT, AgwA, KKH, Lacaton & Vassal, Brunnberg & Forshed, Rodeo Arkitekter, AAA. Page 20-21, references of NL Architects + XVW architectuur, Wallimann Reichen Architects (Rory Gardiner). Page 62-63, images of the authors mentioned.

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Agency for Adaptive Architecture

Works with the existing built environment as a starting point for future development. The practice focuses on adaptive reuse and early-stage strategies that make spatial, technical and economic values visible, comparable and actionable. Founded in 2024 by Henrik Almquist, architect and lecturer at Lund University, AAA develops methods that support informed decisions and long-term value in transformation projects.

Consultancy, research and projects
_ Potential and scenario development
_ Strategic support for municipalities and developers
_ Design and delivery of adaptation and reuse projects

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